Canonical Gravity

Elias Fink

Department of Applied Mathematics and Theoretical Physics,
Centre for Mathematical Sciences,
Wilberforce Road, Cambridge, CB3 OBA, UK

E-mail: ef513@cam.ac.uk

NOTE: These lecture notes are based on the notes I took during the Canonical Gravity course delivered by Malcolm Perry as part of the Part III Mathematics lectures at the University of Cambridge in Lent term 2026. I appreciate any comments or corrections communicated by email. Some useful prerequisites are Classical Mechanics, General Relativity and Quantum Mechanics.

The lecture notes are structured as follows:

- Chapter 1 outlines the fundamentals of Hamiltonian mechanics and constraints as well as applying them to the example of canonical electromagnetism.
- Chapter 2 discusses aspects of foliating a spacetime with hypersurfaces as well as causal structure. We also look at to what extent a set of initial conditions can be used to find solutions to hyperbolic equations of motion.
- Chapter 3 goes over the implementation of classical general relativity into the framework of Hamiltonian mechanics. We apply this to rediscover the De Sitter metric as well as starting to look at the behaviour of spacetime near a singularity.
- Chapter 4 looks at the canonical quantisation of gravity as well as its application to a few issues. Among these are the extraction of thermodynamics from De Sitter and Schwarzschild spacetimes as well as a more advanced look at cosmology and the behaviour near a singularity.

Contents

1 Hamiltonian Mechanics **2**
1.1 Fundamentals of Hamiltonian Mechanics 3
1.1.1 Canonical Transformations and Noether's Theorem 5
1.1.2 A Particle Moving in a Potential 6
1.2 Canonical Quantisation 7
1.2.1 Quantum Harmonic Oscillator 8
1.3 Field Theories 11
1.3.1 Scalar Fields 11
1.3.2 Canonical Electromagnetism 12
1.4 Constraints 14
1.4.1 Canonical Electromagnetism Rebooted 15

2 Foliation and Causal Structure **17**
2.1 Spacelike Hypersurfaces 17
2.2 Foliation 18
2.3 Penrose Diagrams 19
2.4 Initial Value Problem 20

3 Hamiltonian Formulation of Gravity **24**
3.1 Hamiltonian Constraint 26
3.2 Hamiltonian Einstein Equation 28
3.2.1 Cosmology 28
3.2.2 Kasner Universe 29

4 Canonical Quantisation of Gravity **31**
4.1 Wheeler-DeWitt Equation 31
4.1.1 Path Integral Formulation 32
4.2 Quantising De Sitter Spacetime 33
4.2.1 Causal Structure of De Sitter Spacetime 36
4.2.2 Thermodynamics of De Sitter Spacetime 37
4.3 Quantising a Black Hole 39
4.4 Cosmology Rebooted 40
4.5 Behaviour of Singularities 42

1 Hamiltonian Mechanics

The most common approach to studying classical *General Relativity* is the Lagrangian framework. As a metric-based description of gravity, the main object in this framework is the metric tensor g_{ab}, a symmetric rank-2 tensor field defined on a four-dimensional Lorentzian manifold $\mathcal{M}$, called spacetime. We then specify the theory by choosing an action $S[g]$ that is a functional of the metric tensor g_{ab}. For General Relativity, this action is given by the Einstein-Hilbert action

$$S[g_{ab}] = \frac{1}{16\pi G} \int_{\mathcal{M}} d^4x \sqrt{-g}\, R(g_{ab}) \tag{1.1}$$

where we $g = \det(g_{ab})$ is the determinant of the metric, which will be negative for a Lorentzian manifold, and $R(g_{ab})$ is the Ricci scalar associated with the metric tensor. An important observation we can make about this action is diffeomorphism invariance. While we may use a coordinate basis to perform computations using this action, the physics should not depend on it. This means that the action should be invariant under general coordinate transformations, also called diffeomorphisms. Aptly named, the Ricci scalar R is a scalar and hence already manifestly invariant under diffeomorphisms. For a general transformation the measure d^4x picks up a Jacobian factor which is cancelled by the transformation of $\sqrt{-g}$. We therefore refer to the combination $d^4x\sqrt{-g}$ as the invariant measure. This means the action is invariant under such transformations and we can feel more confident in using it to describe our physics.
It is also possible to modify the Einstein-Hilbert action by the addition of a constant term Λ, called the cosmological constant. The action is then

$$S[g_{ab}] = \frac{1}{16\pi G} \int_{\mathcal{M}} d^4x \sqrt{-g}\, \left(R(g_{ab}) - 2\Lambda\right) . \tag{1.2}$$

Applying the principle of least action as outlined below, the Einstein-Hilbert action leads to an equation of motion given by the Einstein field equation

$$R_{ab} - \frac{1}{2} R g_{ab} - \Lambda g_{ab} = 0 \tag{1.3}$$

where R_{ab} is the Ricci tensor. We can add matter terms to the action that give rise to the stress tensor T_{ab}. The Einstein equation is then modified to

$$R_{ab} - \frac{1}{2} R g_{ab} - \Lambda g_{ab} = 8\pi T_{ab} \tag{1.4}$$

where the left hand side of the equation describes the curvature of spacetime and the right hand side describes the energy and momentum distribution. In any coordinate basis, this equation is a set of ten second order coupled PDEs. Finding a solution g_{ab} means determining the gravitational field in this spacetime. We will later discuss in which sense this is possible. Having found the gravitational field, we describe the motion of test particles using another action

$$S[x^a(\lambda)] = \int d\lambda \sqrt{-g_{ab}\frac{dx^a}{d\lambda}\frac{dx^b}{d\lambda}} \tag{1.5}$$

where we are integrating along the trajectory of the particle $x^a(\lambda)$, parametrised by λ.
So far, we have considered General Relativity only in the Lagrangian approach. Canonical Gravity is the study of General Relativity in the framework of Hamiltonian mechanics. This gives access to deep insights about the nature of gravity and hence our universe, that are difficult to reach in a Lagrangian approach. To this end, we would first like to explore the Hamiltonian framework of classical mechanics.

1.1 Fundamentals of Hamiltonian Mechanics

In the Lagrangian approach, a physical system is described by its action functional

$$S[q_i(t)] = \int dt\ L(q_i, \dot{q}_i, t) \tag{1.6}$$

where t acts as a time parameter. The state of the system is given by the dynamical variables q_i and their t-derivatives $\dot{q}_i$. The integrand of the action functional is known as the Lagrangian L. The equations of motion are determined by applying the principle of least action which states that the classical path taken is the one that extremises the action or mathematically

$$\delta S = 0 \tag{1.7}$$

where δ denotes a functional variation of the path $\{q_i(t)\}$, keeping the end points fixed. We can now use the integral form of the action functional to derive the equations of motion as

$$S' = S + \delta S = \int dt\ L(q_i + \delta q_i, \dot{q}_i + \delta\dot{q}_i, t) = \int dt\ L(q_i, \dot{q}_i, t) + \frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\delta\dot{q}_i \tag{1.8}$$

and hence

$$\delta S = \int dt\ \frac{\partial L}{\partial q_i}\delta q_i + \frac{\partial L}{\partial \dot{q}_i}\delta\dot{q}_i = \int dt\ \left[\frac{\partial L}{\partial q_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i}\right]\delta q_i \tag{1.9}$$

where we have integrated by parts and no boundary terms appear due to holding the end points fixed. This means that for a general variation δq_i, the principle of least action is equivalent to the Euler-Lagrange equation

$$\frac{\partial L}{\partial q_i} - \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = 0 \tag{1.10}$$

which generally gives a system of coupled second order differential equations that can be solved analytically or numerically to determine the path taken by the classical system subject to boundary conditions. We would like to take a different approach now.
We first define a canonical momentum

$$p_i = \frac{\partial L}{\partial \dot{q}_i} \tag{1.11}$$

for each degree of freedom and perform a Legendre transform from real space to phase space $(q_i, \dot{q}_i) \to (q_i, p_i)$ to find the Hamiltonian

$$H(q_i, p_i, t) = \sum_i \dot{q}_i p_i - L(q_i, \dot{q}_i, t)\ . \tag{1.12}$$

This will in general involve inverting (1.11) to find $\dot{q}_i$ as a function of q_i and p_i in order to eliminate it in the expression for the Hamiltonian $H(q_i, p_i, t)$. In practise, it may be difficult to do this in which case the Hamiltonian framework could be unsuitable for performing calculations on that particular system.

Classically, the Hamiltonian is a measure of the total amount of energy in the system. One useful property is that it is conserved for systems where the Lagrangian does not explicitly depend on time. This can be seen by

$$\frac{dH}{dt} = \ddot{q}_i p_i + \dot{q}_i p_i - \frac{\partial L}{\partial t} - \frac{\partial L}{\partial q_i}\dot{q}_i - \frac{\partial L}{\partial \dot{q}_i}\ddot{q}_i = 0 \tag{1.13}$$

where we have adopted Einstein summation convention to drop the explicit sum and used the Euler-Lagrange equation. Mathematically, this property stems from Noether's theorem where H acts as the conserved charge associated with time translation symmetry.

The Hamiltonian can also be used to determine the equations of motion directly through Hamilton's equations

$$\dot{q}_i = \frac{\partial H}{\partial p_i},\ \dot{p}_i = -\frac{\partial H}{\partial q_i} \tag{1.14}$$

which is equivalent to

$$\dot{q}_i = \{q_i, H\},\ \dot{p}_i = \{p_i, H\}\ . \tag{1.15}$$

The Poisson brackets $\{A, B\}$ are defined as

$$\{A, B\} = \frac{\partial A}{\partial q_i}\frac{\partial B}{\partial p_i} - \frac{\partial B}{\partial q_i}\frac{\partial A}{\partial p_i} \tag{1.16}$$

where we have again used Einstein notation to sum over the repeated index i. This can be used to show that for any analytic function of the phase space variables $f = f(q_i, p_i)$, the time derivative can be found as

$$\dot{f} = \{f, H\} \tag{1.17}$$

and hence H is the generator of time translations. We may also use this to show that any Hamiltonian $H = H(q_i, p_i)$ not explicitly dependent on time satisfies

$$\dot{H} = \{H, H\} = 0 \tag{1.18}$$

and is therefore conserved. Hamiltonian mechanics using Hamilton's equations is equivalent to Lagrangian mechanics using the principle of least action as

$$\dot{q}_i = \frac{\partial H}{\partial p_i} = \dot{q}_i,\ \frac{d}{dt}\frac{\partial L}{\partial \dot{q}_i} = \dot{p}_i = -\frac{\partial H}{\partial q_i} = \frac{\partial L}{\partial q_i} \tag{1.19}$$

where we can see that, given the definition of the Hamiltonian in terms of a Legendre transform, the first Hamilton's equation becomes trivial and the second reproduces the Euler-Lagrange equation.

1.1.1 Canonical Transformations and Noether's Theorem

This section is not part of the Canonical Gravity course but is included for the reader's benefit as an adapted version from the Advanced Classical Physics course taught by Andrew Tolley at Imperial College London.

For a general function $f = f(q_i, p_i, t)$, the time evolution is given by

$$\frac{df}{dt} = \frac{\partial f}{\partial t} + \{f, H\}_{q_i,p_j} \tag{1.20}$$

where we have explicitly noted the Poisson bracket with respect to the phase space parameters $\{q_i\}, \{p_i\}$. In the spirit of coordinate invariance however, we should be able to re-parametrise phase space while still describing the same physical system. We therefore define another set of coordinates $Q_I(q_i, p_i, t)$ and $P_J(q_i, p_i, t)$ on phase space. As all dynamics are controlled by the Poisson brackets, we require the new coordinates to satisfy

$$\{f, g\}_{Q_I,P_I} = \{f, g\}_{q_i,p_i} \tag{1.21}$$

for arbitrary phase space functions f, g. Any such transformation is known as a canonical transformation. It is possible to show that to satisfy (1.21), it is sufficient to require

$$\{Q_I, Q_J\}_{q_i,p_i} = \{P_I, P_J\}_{q_i,p_i} = 0,\ \{Q_I, P_J\}_{q_i,p_i} = \delta_{IJ} \tag{1.22}$$

where any pairs Q_i and P_i are known as canonical conjugates.

Back in the Lagrangian picture, the most general transformation we can make is one that changes the Lagrangian by a total derivative to keep the equations of motion invariant. Take the new form of the Hamiltonian after a canonical transformation to be $K(Q_I, P_I, t)$. The new Lagrangian can then be written as

$$L' = P_I \dot{Q}_I - K(Q_I, P_I, t) \tag{1.23}$$

but we also know that this has to differ from the original Lagrangian by no more than a total derivative. Therefore we obtain the relation

$$P_I \dot{Q}_I - K(Q_I, P_I, t) = p_i \dot{q}_i - H(q_i, p_i, t) + \frac{dF}{dt} \tag{1.24}$$

where $F = F(q_i, p_i, t)$ is an arbitrary analytic function. Applying the chain rule, we may expand the relation as

$$P_I \frac{\partial Q_I}{\partial t} + P_I \frac{\partial Q_I}{\partial q_i} \dot{q}_i + P_I \frac{\partial Q_I}{\partial p_i} \dot{p}_i - K = p_i \dot{q}_i - H + \frac{\partial F}{\partial t} + \frac{\partial F}{\partial q_i} \dot{q}_i + \frac{\partial F}{\partial p_i} \dot{p}_i\ . \tag{1.25}$$

As we still see each q_i and p_i as independent coordinates on phase space, we can compare these coefficients in each of the terms to split the relation into three conditions

$$P_I \frac{\partial Q_I}{\partial q_i} = p_i + \frac{\partial F}{\partial q_i} \tag{1.26}$$

$$P_I \frac{\partial Q_I}{\partial p_i} = \frac{\partial F}{\partial p_i} \tag{1.27}$$

$$K = H - P_I \frac{\partial Q_I}{\partial t} - \frac{\partial F}{\partial t} \tag{1.28}$$

satisfied by all valid canonical transformations. This describes the most general form of a canonical transformation. Let's now consider one particular application of infinitesimal transformations of the form

$$Q_i = q_i + \delta q_i,\ P_i = p_i + \delta p_i,\ F = A\delta\lambda,\ K = H + \delta H \tag{1.29}$$

where we only keep terms to first order in δ. This means the first-order contribution to the conditions reduces to

$$\delta p_i + p_j \frac{\partial \delta q_j}{\partial q_i} = \frac{\partial A}{\partial q_i}\delta\lambda \tag{1.30}$$

$$p_j \frac{\partial \delta q_j}{\partial p_i} = \frac{\partial A}{\partial p_i}\delta\lambda \tag{1.31}$$

$$\delta H = p_j \frac{\partial \delta q_j}{\partial t} - \frac{\partial A}{\partial t}\delta\lambda\ . \tag{1.32}$$

From these variables, we now define another function

$$G = p_j \frac{\delta q_j}{\delta\lambda} - A \tag{1.33}$$

which we may recognise as the definition of the Noether charge when considering symmetry transformations of an action. To make this more clear, G allows us to rewrite the equations as

$$\delta q_i = \frac{\partial G}{\partial p_i}\delta\lambda = \{q_i, G\}\delta\lambda \tag{1.34}$$

$$\delta p_i = -\frac{\partial G}{\partial q_i}\delta\lambda = \{p_i, G\}\delta\lambda \tag{1.35}$$

$$\delta H = \frac{\partial G}{\partial t}\delta\lambda \tag{1.36}$$

where we can now clearly see that if the Hamiltonian is invariant under the transformation then G is conserved. The phase space variables also infinitesimally transform in a way we would expect from the generator of a symmetry. We have reached the result of Noether's theorem expressed in the Hamiltonian framework.

1.1.2 A Particle Moving in a Potential

The discussion of Lagrangian and Hamiltonian mechanics has so far been fairly abstract. To illustrate how to apply this to a specific system, we pick a particle of mass m moving in a position-dependent potential $V(x_i)$ in d-dimensional space. The corresponding action functional is

$$S[x_i(t)] = \int dt\ \frac{1}{2}m\dot{x}_i^2 - V(x_i) \tag{1.37}$$

where the degrees of freedom of the system $\{x_i\}$ are now components of a d-dimensional position vector. The Euler-Lagrange equation of this system evaluates to

$$m\ddot{x}_i = -\frac{\partial V}{\partial x_i} \tag{1.38}$$

which is a system of coupled second order differential equations given a general potential. Another approach is to find the canonical momenta $p_i = m\dot{x}_i$ and hence the Hamiltonian

$$H = \frac{p_i^2}{2m} + V(x_i) \tag{1.39}$$

which can be used to find Hamilton's equations

$$\dot{x}_i = \frac{p_i}{m},\ \dot{p}_i = -\frac{\partial V}{\partial x_i} \tag{1.40}$$

that are indeed equivalent to the Euler-Lagrange equation. We also notice that both the Euler-Lagrange equation and Hamilton's equation are equivalent to the result that would have been obtained from treating a particle moving in a potential in the Newtonian approach

$$\mathbf{F} = -\nabla V = m\mathbf{a} = m\frac{d^2\mathbf{x}}{dt^2} \tag{1.41}$$

by considering the forces acting on it and applying Newton's second law. This is important sanity check since all three are simply different mathematical formulations of the same underlying physics and are hence expected to yield the same behaviour.

1.2 Canonical Quantisation

A feature of the Hamiltonian approach is that it enables a simple framework for quantising a classical theory. When canonically quantising a system with degrees of freedom $\{q_i\}$, these variables are promoted to linear operators $\{\hat{q}_i\}$ acting on states in a Hilbert space. These operators obey commutation relations that can be obtained from Poisson brackets

$$\{A, B\} \to \frac{1}{i\hbar}[\hat{A}, \hat{B}] \tag{1.42}$$

where all variables and functions are promoted to linear operators on a Hilbert space accordingly. Using the example of the particle in a potential, we can compute the commutator between position and momentum operators as

$$\{x_i, p_j\} = \delta_{ij} \to [\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij}\ . \tag{1.43}$$

This map motivates the introduction of the Heisenberg picture where states in the Hilbert space are taken to be constant and all the time dependence is instead put onto the operators. This is possible since the only measurable property about a quantum theory are its overlap amplitudes

$$\langle\phi|\hat{A}|\psi\rangle \tag{1.44}$$

for any states $|\phi\rangle, |\psi\rangle$ and any self-adjoint operator $\hat{A}$ corresponding to a physical observable. In the commonly used Schrödinger picture, the operator is taken to be constant and the states to unitarily evolve as

$$|\phi\rangle \to |\phi(t)\rangle = U(t)\,|\phi(0)\rangle = e^{-\frac{i}{\hbar}\hat{H}t}\,|\phi(0)\rangle \tag{1.45}$$

where $\hat{H}$ is the Hamiltonian operator. When computing the overlap

$$\langle\phi(t)|\hat{A}|\psi(t)\rangle = (\langle\phi(0)|e^{\frac{i}{\hbar}\hat{H}t})\hat{A}(e^{-\frac{i}{\hbar}\hat{H}t}|\psi(0)\rangle) = \langle\phi(0)|(e^{\frac{i}{\hbar}\hat{H}t}\hat{A}e^{-\frac{i}{\hbar}\hat{H}t})|\psi(0)\rangle = \langle\phi|\hat{A}(t)|\psi\rangle \tag{1.46}$$

it can be seen that the time evolution can be factored onto the operator to move from Schrödinger picture, where states evolve according to the Schrödinger equation, to the Heisenberg picture, where the operators evolve according to the Heisenberg equation

$$\frac{d\hat{A}}{dt} = -\frac{i}{\hbar}[\hat{A}, \hat{H}] \ . \tag{1.47}$$

The solution of this equation reproduces the expected time evolution of $\hat{A}$ in (1.46) and is also consistent with the quantisation map (1.42).

1.2.1 Quantum Harmonic Oscillator

A simple example to test the power of canonical quantisation is a particle of mass m moving in a quadratic potential in d dimensions. Classically, this is described by the action

$$S = \int dt\, \frac{1}{2}m\dot{x}_i^2 - \frac{1}{2}m\omega^2 x_i^2 \tag{1.48}$$

where ω is a constant with units of frequency. We may easily compute the classical equations of motion

$$\ddot{x}_i = -\omega^2 x_i \tag{1.49}$$

resulting in d independent second order ODEs. To turn the problem into a Hamiltonian one, we find the canonical momenta $p_i = m\dot{x}_i$ and the Hamiltonian

$$H = \frac{p_i^2}{2m} + \frac{1}{2}m\omega^2 x_i^2 \tag{1.50}$$

giving Hamilton's equations

$$\dot{x}_i = \frac{p_i}{m},\ \dot{p}_i = -m\omega^2 x_i \ . \tag{1.51}$$

The Hamiltonian picture now makes it easy to find a quantum theory that corresponds to a harmonic oscillator in the classical limit. To do this, we promote the coordinates x_i and momenta p_i to linear operators $\hat{x}_i, \hat{p}_i$. The Poisson brackets are telling us that we should define these to obey the commutation relations

$$[\hat{x}_i, \hat{x}_j] = [\hat{p}_i, \hat{p}_j] = 0,\ [\hat{x}_i, \hat{p}_j] = i\hbar\delta_{ij} \tag{1.52}$$

and as all operators in the theory are built out of $\hat{x}_i$ and $\hat{p}_i$, we have hence specified all commutation relations. The Hamiltonian operator is given by

$$\hat{H} = \frac{\hat{p}_i^2}{2m} + \frac{1}{2}m\omega^2\hat{x}_i^2 \ . \tag{1.53}$$

Having constructed the Hamiltonian operator, we can use the Schrödinger or Heisenberg equation to figure out how any state of the system will evolve over time. But what does the Hilbert space look like? To determine this, we define new operators

$$\hat{a}_i = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x}_i + \frac{i}{m\omega}\hat{p}_i\right) \tag{1.54}$$

$$\hat{a}_i^\dagger = \sqrt{\frac{m\omega}{2\hbar}}\left(\hat{x}_i - \frac{i}{m\omega}\hat{p}_i\right) \tag{1.55}$$

referred to as the lowering and raising ladder operators and obeying the commutation relation

$$[\hat{a}_i, \hat{a}_j^\dagger] = \delta_{ij} \ . \tag{1.56}$$

The Hamiltonian now factorises as

$$\hat{H} = \hbar\omega\left(\hat{a}_i^\dagger \hat{a}_i + \frac{d}{2}\right) \tag{1.57}$$

where the final constant is understood to multiply an identity operator. The ground state of the system is the state with the lowest energy. Quantum mechanically we would like to use the state that gives the lowest expectation value for the Hamiltonian. For a general state $|\psi\rangle$, we have the identity

$$\langle\psi|\hat{a}_i^\dagger \hat{a}_i|\psi\rangle = \sum_i ||\hat{a}_i |\psi\rangle ||^2 \geq 0 \tag{1.58}$$

and so the ground state should satisfy $\hat{a}_i |0\rangle = 0$ for all i. This ground state is an energy eigenstate

$$\hat{H} |0\rangle = \frac{d}{2}\hbar\omega \tag{1.59}$$

and we can reach other energy eigenstates by applying raising operators $\hat{a}_i^\dagger |0\rangle$. Applying the commutation relations we see that applying raising and lowering operators

$$\hat{H}(\hat{a}_i^\dagger |\psi\rangle) = (\hat{a}_i^\dagger \hat{H} + \hbar\omega) |\psi\rangle \ , \ \hat{H}(\hat{a}_i |\psi\rangle) = (\hat{a}_i \hat{H} - \hbar\omega) |\psi\rangle \tag{1.60}$$

raises or lowers the energy of the state by quanta of $\hbar\omega$. A general energy eigenstate of the harmonic oscillator is therefore given by applying a sequence of raising operators

$$(\hat{a}_1^\dagger)^{n_1} ... (\hat{a}_d^\dagger)^{n_d} |0\rangle = |n_1, ..., n_d\rangle \tag{1.61}$$

and has energy

$$\hat{H} |n_1, ...n_d\rangle = \hbar\omega \sum_i \left(n_i + \frac{1}{2}\right) \tag{1.62}$$

which fully specifies the Hilbert space and spectrum of the theory. Having explored this simple oscillator model, it is easy to modify it to more general setups. For example we may wish to have a different constant ω_i^2 multiply the potential for each direction x_i (where we have suspended summation notation). This preserves the structure of the theory but modifies the energy of eigenstates to

$$\hat{H}\left|n_1,...n_d\right\rangle = \hbar\sum_i\left(\omega_i n_i + \frac{1}{2}\right) . \tag{1.63}$$

Another even more generalised version of the model mixes the dimensions with the potential

$$V(x_i) = \frac{1}{2}M_{ij}x_i x_j \tag{1.64}$$

which can arise as a quadratic approximation around a local extremum of the potential landscape of some more complicated theory. Note that the potential matrix M_{ij} will always be diagonalisable since the elements are real and contracting with $x_i x_j$ picks out the symmetric part. If M_{ij} has negative eigenvalues then the Hamiltonian is unbounded below and hence we cannot make sense of the quantum theory as the system will simply display runaway behaviour. If M_{ij} has zero eigenvalues then these directions behave like free particles and so the wave function spreads out from the initial distribution. This can give a sensible quantum theory but we do not wish to discuss it here. The third case is that M_{ij} only has positive eigenvalues. In this case we perform an orthogonal transformation $R \in O(d)$ on x_i that leaves the kinetic term invariant but brings the potential term into a diagonalised form. As we learnt before, we must check the Poisson brackets to make sure this corresponds to a canonical transformation. These are

$$\{R_{ik}x_k, R_{jl}x_l\} = R_{ik}R_{jl}\{x_k, x_l\} = 0,\ \{p_i, p_j\} = 0,\ \{R_{ik}x_k, p_j\} = R_{ik}\{x_k, p_j\} = R_{ij} \tag{1.65}$$

where we have used the linearity of the Poisson bracket. Hmm, this is not quite right. The canonical momenta should transform with the same transformation as the coordinates. Plugging this in gives

$$\{R_{ik}x_k, R_{jl}p_l\} = R_{ik}R_{jl}\{x_k, p_l\} = R_{ik}R_{jk} = (RR^T)_{ij} = \delta_{ij} \tag{1.66}$$

since we have chosen R to be orthogonal. After performing this transformation, we are back to the case with different constants ω_i^2 that now correspond to the eigenvalues of M_{ij}. The final check we would like to make is whether the quantum theory does actually reduce to the classical theory in the classical limit. For this we will need the commutators

$$[\hat{x}_i, \hat{H}] = i\hbar\frac{\hat{p}_i}{m},\ [\hat{p}_i, \hat{H}] = -i\hbar m\omega_i^2\hat{x}_i \tag{1.67}$$

and work in the Heisenberg picture. In any state, the expectation values now satisfy

$$\frac{d\left\langle\hat{x}_i\right\rangle}{dt} = -\frac{i}{\hbar}\left\langle[\hat{x}_i, \hat{H}]\right\rangle = \frac{\left\langle\hat{p}_i\right\rangle}{m},\ \frac{d\left\langle\hat{p}_i\right\rangle}{dt} = -\frac{i}{\hbar}\left\langle[\hat{p}_i, \hat{H}]\right\rangle = -m\omega_i^2\left\langle\hat{x}_i\right\rangle \tag{1.68}$$

which are indeed Hamilton's equations for this system. As the expectation values of quantum observable become what we classically understand as their values, this confirms that in the classical limit the canonically quantised theory reduces to its classical description. The result (1.68) is sometimes known as the Ehrenfest theorem and can easily be shown to hold for general Hamiltonian theories.

1.3 Field Theories

General Relativity is a field theory and so we would like to find out how to implement Hamiltonian mechanics for classical fields. All actions we construct should be compatible with special relativity and we will see shortly what constraints this puts on the theories we can write down.

1.3.1 Scalar Fields

The simplest type of field theory is a scalar theory. A scalar field is simply a function $f = f(x)$ where x refers to all dimensions of spacetime. Consider a functional $F[f(x)]$ and define the functional derivative

$$\frac{\delta F}{\delta f(x')} = \lim_{\epsilon \to 0} \frac{F[f(x) + \epsilon \delta(x - x')] - F[f(x)]}{\epsilon} \tag{1.69}$$

similarly to a function derivative. Now consider an action of a single scalar field in d dimensions of spacetime

$$S[\phi] = \int dt\, L[\phi, \dot{\phi}, t] = \int dt\, d^{d-1}x\, \mathcal{L}(\phi, \partial\phi, t) \tag{1.70}$$

where L is the Lagrangian and the new object $\mathcal{L}$ is the Lagrangian density. There are now important conditions on $\mathcal{L}$ for the theory to be compatible with relativity. Most importantly, $\mathcal{L}$ must be local. This means that it may only depend on the fields and their derivatives evaluated at a single point x instead of mixing contributions from different points of spacetime. The other requirement is that the Lagrangian density as well as the integral measure are Lorentz invariant. Practically this means that we should not have any lose tensor (or spinor) indices hanging around and everything should be paired in a way that any factors arising from Lorentz transformations are cancelled. Once we have made sure that this is satisfied we define the conjugate momentum to ϕ as

$$\pi(x) = \frac{\delta \mathcal{L}}{\delta \dot{\phi}(x)} \tag{1.71}$$

where $\dot{}$ still refers to derivatives with respect to t. We also define the field theory Poisson bracket

$$\{X(x), Y(x')\} = \int d^{d-1}x'' \left[\frac{\delta X(x)}{\delta \phi(x'')} \frac{\delta Y(x')}{\delta \pi(x'')} - \frac{\delta Y(x')}{\delta \phi(x'')} \frac{\delta X(x)}{\delta \pi(x'')} \right] \tag{1.72}$$

which leads to the canonical Poisson bracket

$$\{\phi(x), \pi(x')\} = \delta(x - x') \tag{1.73}$$

treating ϕ and π as independent fields. Having defined Poisson brackets, we are able define the Hamiltonian

$$H[\phi, \pi, t] = \int d^{d-1}x\, \dot{\phi}\pi - \mathcal{L} \tag{1.74}$$

to state Hamilton's equations for fields

$$\dot{\phi} = \{\phi, H\}, \; \dot{\pi} = \{\pi, H\} \; . \tag{1.75}$$

To demonstrate how the Hamiltonian approach is applied for field theories, we consider the action of a free scalar field in 4d Minkowski spacetime

$$S = \int dt\, d^3x \; -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - \frac{1}{2}m^2\phi^2 = \int dt\, d^3x \; \frac{1}{2}\dot{\phi}^2 - \frac{1}{2}(\nabla\phi)^2 - \frac{1}{2}m^2\phi^2 \tag{1.76}$$

where we employ the $(-,+,+,+)$ metric signature. We hence find $\pi = \dot{\phi}$ and

$$H = \int d^3x \; \frac{1}{2}\pi^2 + \frac{1}{2}(\nabla\phi)^2 + \frac{1}{2}m^2\phi^2 \tag{1.77}$$

where we note the Hamiltonian is non-negative. This is good since there is a lower bound to the energy $H \geq 0$ and hence no runaway behaviour. We may now apply Hamilton's equations to find

$$\dot{\phi} = \{\phi, H\} = \pi, \; \dot{\pi} = \{\pi, H\} = -m^2\phi + \nabla^2\phi \tag{1.78}$$

where the first relation is trivially the definition of the canonical momentum and the second reproduces the canonical form of the Klein-Gordon equation, the expected equation of motion for a massive scalar field. One drawback of using the Hamiltonian framework for any relativistic calculations is that it is not manifestly Lorentz invariant. To define the canonical momentum and Hamiltonian, we had to pick out one time direction which explicitly breaks Lorentz invariance. Once we follow through with the calculation, the result should still be compatible with relativity as Hamiltonian mechanics is equivalent to the manifestly Lorentz invariant Lagrangian approach to field theory. To see this, we substitute the field Hamilton's equation into the second to find

$$\ddot{\phi} - \nabla^2\phi + m^2\phi = (\Box + m^2)\phi = 0 \tag{1.79}$$

which is manifestly Lorentz invariant. When constructing a canonical description of gravity we will run into a similar issue where we are forced to explicitly break diffeomorphism invariance by foliating spacetime and only recover it with in the final result.
Naturally, the Hamiltonian considerations on scalar fields can again be used to perform canonical quantisation. This leads to subject of Quantum Field Theory which has gained some popularity over the years. This, however, is beyond the scope of what we want to do here.

1.3.2 Canonical Electromagnetism

Having considered the most simple field theory, we are able to explore a more interesting field theory describing electromagnetism. Similarly to gravity, Maxwell theory is a gauge theory and there are many features of canonical electromagnetism we will recognise when we go on to study gravity in this framework. We start by writing down the Maxwell action

$$S[A_\mu] = -\frac{1}{4}\int d^4x \; F_{\mu\nu}F^{\mu\nu} \tag{1.80}$$

with the field strength defined in the usual way

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu \tag{1.81}$$

where A_μ is the gauge potential. The action is invariant under the gauge transformation $A_\mu \to A_\mu + \partial_\mu \lambda$. Varying the action gives the equation of motion

$$\partial_\mu F^\mu{}_\nu = 0 \tag{1.82}$$

together with the Bianchi identity

$$\partial_{[\mu} F_{\nu\rho]} = 0 \tag{1.83}$$

which is satisfied trivially through (1.81). The gauge potential and gauge-invariant field strength are related to the physical electric and magnetic fields by

$$E_i = F_{i0} = \dot{A}_i - \partial_i A_0, \; B_i = \frac{1}{2}\epsilon_{ijk} F_{jk} = \epsilon_{ijk} \partial_j A_k \; . \tag{1.84}$$

The action can now be rewritten as

$$\begin{aligned} S &= \int dt\, d^3x \; \frac{1}{2} F_{0i} F_{0i} - \frac{1}{4} F_{ij} F_{ij} \\ &= \int dt\, d^3x \; \frac{1}{2} \dot{A}_i^2 - (\partial_i A_0) \dot{A}_i + (\partial_i A_0)^2 - \frac{1}{2} \partial_i A_j \partial_i A_j + \frac{1}{2} \partial_j A_i \partial_i A_j \end{aligned} \tag{1.85}$$

where it is easy to determine the conjugate momenta

$$\pi_i = \dot{A}_i - \partial_i A_0 = E_i \tag{1.86}$$

and $\pi_0 = 0$. It may be slightly concerning that one of the conjugate momenta is zero but we push on for now. As we will see this is a common feature of gauge theories. We can propose the naïve Hamiltonian

$$H = \int d^3x \; \pi_i \dot{A}_i - \mathcal{L} = \int d^3x \; \frac{1}{2} \pi_i^2 + \frac{1}{2} B_i^2 + \pi_i \partial_i A_0 \tag{1.87}$$

however this cannot be fully correct since we have not enforced the primary constraint $\pi_0 = 0$. We do this using a Lagrange multiplier ν_1 and propose

$$H = \int d^3x \; \frac{1}{2} \pi_i^2 + \frac{1}{2} B_i^2 - A_0 \partial_i \pi_i + \nu_1 \pi_0 \tag{1.88}$$

where we have integrated by parts on the third term. This does not work again since now

$$\dot{\pi}_0 = -\frac{\delta H}{\delta A_0} = \partial_i \pi_i \tag{1.89}$$

which is not generally zero. This gives a secondary constraint $\partial_i \pi_i = 0$, known as Gauss' law, that we will again fix using a Lagrange multiplier ν_2. We can now finally propose the correct Hamiltonian

$$H = \int d^3x \; \frac{1}{2} \pi_i^2 + \frac{1}{2} B_i^2 + (\nu_2 - A_0) \partial_i \pi_i + \nu_1 \pi_0 \tag{1.90}$$

where we can now use the equations of motion to determine the Lagrange multipliers. Using the Hamilton's equations for A_0 and A_i tells us that we can fix $\nu_1 = \dot{A}_0$ and $\nu_2 = 0$. Hence the Hamiltonian

$$H = \int d^3x \, \frac{1}{2}\pi_i^2 + \frac{1}{2}B_i^2 - A_0\partial_i\pi_i + \dot{A}_0\pi_0 \tag{1.91}$$

reproduces the correct Maxwell equations. We could now go on to fix the gauge freedom but first we will take an excursion to the land of constraints to find a more systematic way of constraining the Hamiltonian correctly.

1.4 Constraints

When constructing Hamiltonians as in the case of electromagnetism, we may find the need to impose constraints on the system. We will label these primary constraints as $\phi_\alpha = 0$ where ϕ_α is an analytic function of the degrees of freedom of the system and we require the constraints to be linearly independent. Consider an unconstrained system with phase space variables $\{q_i\}, \{p_i\}$ and Hamiltonian $H(q_i, p_i)$. The constrained Hamiltonian is constructed by the addition of constraint terms

$$\tilde{H}(q_i, p_i) = H(q_i, p_i) + \nu_\alpha \phi_\alpha(q_i, p_i) \tag{1.92}$$

where ν_α are the Lagrange multipliers. An equivalent definition also applied to field theories where the implicit sum is exchanged for an integral. The constrained Hamiltonian changes the Poisson brackets to

$$\dot{q}_i = \{q_i, \tilde{H}\} = \frac{\partial H}{\partial p_i} + \nu_\alpha \frac{\partial \phi_\alpha}{\partial p_i}, \; \dot{p}_i = \{p_i, \tilde{H}\} = -\frac{\partial H}{\partial q_i} - \nu_\alpha \frac{\partial \phi_\alpha}{\partial q_i} \tag{1.93}$$

as well as introducing new Poisson brackets

$$\dot{\phi}_\alpha = \{\phi_\alpha, \tilde{H}\} = \{\phi_\alpha, H\} + \nu_\beta\{\phi_\alpha, \phi_\beta\} = 0 \tag{1.94}$$

which give either conditions on ν_α or new secondary constraints. This process is repeated until there are no new constraints introduced by the existing constraints.
Consider any analytic function $A(q_i, p_i)$ and its Poisson bracket $\{A, \phi_\alpha\}$ with the constraints. If this Poisson bracket is zero for all α then A is considered first class. If not then A is second class. As constraints are always analytic functions of the phase space variables, they can themselves be classified into first class ψ_α and second class χ_α constraints. For second class constraints we define

$$\{\chi_\alpha, \chi_\beta\} = C_{\alpha\beta} \; . \tag{1.95}$$

We require the determinant of $C_{\alpha\beta}$ to be non-zero since otherwise there are combinations of second class constraints that are actually first class and hence the set $\{\chi_\alpha\}$ is reducible. Therefore, we are able to introduce the prime operation $'$ that acts on functions A as

$$A \to A' = A - \{A, \chi_\alpha\}C^{-1}_{\alpha\beta}\chi_\beta \tag{1.96}$$

such that

$$\{A', \chi_\gamma\} = \{A, \chi_\gamma\} - \{A, \chi_\alpha\} C^{-1}_{\alpha\beta} \{\chi_\beta, \chi_\gamma\} = 0 \; . \tag{1.97}$$

This motivates us to define the Dirac bracket

$$\{A, B\}_D = \{A', B\} = \{A, B'\} = \{A', B'\} = \{A, B\} - \{A, \chi_\alpha\} C^{-1}_{\alpha\beta} \{\chi_\beta, B\} \tag{1.98}$$

which can be shown to satisfy

$$\{C, \chi_\gamma\}_D = 0 \tag{1.99}$$

for any analytic function $C(q_i, p_i)$. Now take the primed Hamiltonian

$$H' = H - \{H, \chi_\alpha\} C^{-1}_{\alpha\beta} \chi_\beta \tag{1.100}$$

which automatically satisfies

$$\dot{\chi}_\alpha = \{\chi_\alpha, H'\} = 0 \tag{1.101}$$

for all second class constraints χ_α. Hence we can identify H' with $\tilde{H}$ to find that

$$\nu_\beta = -\{H, \chi_\alpha\} C^{-1}_{\alpha\beta} \tag{1.102}$$

for second class constraints. We have now shown that the Lagrange multipliers can be determined for second class constraints by the requirement that constraints stay satisfied. For a general function A we can now find that

$$\dot{A} = \{A, H\}_D \tag{1.103}$$

gives the time evolution subject to second class constraints. Now let us turn our eyes to first class constraints. These are usually associated with fixing gauge freedom. We introduce the gauge conditions $G_\alpha = 0$. Looking at second class constraints, we decide that a choice of gauge is sensible if

$$\det(\{\phi_\alpha, G_\beta\}) \neq 0 \tag{1.104}$$

such that we can treat the set of constraints $\{\phi_\alpha\} = \{\chi_\alpha\} \cup \{\psi_\alpha\} \cup \{G_\alpha\}$ on equal footing as second class constraints. It is important to emphasise that for this to work, we must choose G_α to satisfy (1.104).

1.4.1 Canonical Electromagnetism Rebooted

Armed with the knowledge of constraints, we can now look at electromagnetism through this lens. At first, we only have the primary constraint $\pi_0 = 0$ which satisfies $\{\pi_0(x), \pi_0(x')\} = 0$ and so is first class. The secondary constraint $\dot{\pi}_0 = 0$ requires $\partial_i \pi_i = 0$ which can also be shown to be first class. The gauge choice we will make is radiation gauge $A_0 = 0,\ \partial_i A_i = 0$. These contribute to the fully constrained Hamiltonian

$$H = \int d^3x \, \frac{1}{2}\pi_i^2 + \frac{1}{2} B_i^2 + \pi_i \partial_i A_0 + \nu_1 \pi_0 + \nu_2 \partial_i \pi_i + \nu_3 A_0 + \nu_4 \partial_i A_i \; . \tag{1.105}$$

Treating all constraints as second class constraints, we can find the Poisson matrix

$$C_{\alpha\beta} = \begin{pmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & \partial_x^2 \\ -1 & 0 & 0 & 0 \\ 0 & -\partial_x^2 & 0 & 0 \end{pmatrix} \delta(x - x') \tag{1.106}$$

which is invertible. Using the Green's function of the Laplacian operator as inverse, we find

$$C^{-1}_{\alpha\beta} = \begin{pmatrix} 0 & 0 & -\delta(x-x') & 0 \\ 0 & 0 & 0 & \frac{1}{4\pi|x-x'|} \\ \delta(x-x') & 0 & 0 & 0 \\ 0 & \frac{-1}{4\pi|x-x'|} & 0 & 0 \end{pmatrix} . \tag{1.107}$$

This tells us that the gauge choice of radiation gauge is a sensible one. Now let us see what it tells us about the Lagrange multipliers. For ν_1 we find

$$\nu_1 = -\{H, \int d^3x \, \chi_3(x)\} C^{-1}_{31} = \nu_1 \tag{1.108}$$

so no new information in this case. This is also true for the other Lagrange multipliers. We can also compute the primed Hamiltonian

$$\begin{aligned} H' = H - \{H, \chi_1\} C^{-1}_{13} \chi_3 - \{H, \chi_2\} C^{-1}_{24} \chi_4 - \{H, \chi_3\} C^{-1}_{31} \chi_1 - \{H, \chi_4\} C^{-1}_{42} \chi_2 \\ = \int d^3x \, \frac{1}{2}\pi_i^2 + \frac{1}{2} B_i^2 \end{aligned} \tag{1.109}$$

which is fully compatible with the constraints A_0, π_0, $\partial_i A_i$ and $\partial_i \pi_i$. The constrained equations of motion are now again implemented using Dirac brackets.

2 Foliation and Causal Structure

Studying the Hamiltonian approach to field theories in Minkowski spacetime, we learnt that we have to perform a 3+1 decomposition, picking a time direction, to be able to determine the Hamiltonian. While this approach explicitly breaks Lorentz invariance, the results will turn out not to depend on what time function we picked and are hence Lorentz invariant.

2.1 Spacelike Hypersurfaces

Consider a four-dimensional Lorentzian spacetime $(\mathcal{M}, g)$ and pick a function $f(x^a)$ such that $f = 0$ describes a 3-surface with timelike unit normal $n_a = \lambda \nabla_a f$ where ∇ is the Levi-Civita connection on the manifold. Therefore the set of points where $f = 0$ describe a spacelike hypersurface Σ. The induced metric (or first fundamental form) on the hypersurface

$$h_{ab} = g_{ab} + n_a n_b \tag{2.1}$$

is Riemannian as no two points on the hypersurface can be connected by a timelike or null curve. It is now possible to decompose any vector V^a into

$$V^a = g^a_b V^b = h^a_b V^b - n^a n_b V^b \tag{2.2}$$

a piece in Σ and a piece normal to Σ. We may hence treat the induced metric h^a_b as a projection operator into the hypersurface. To this end we define the extrinsic curvature (or second fundamental form)

$$K_{ab} = -\frac{1}{2}\mathcal{L}_n h_{ab} = h^c{}_a h^d{}_b \nabla_c n_d \tag{2.3}$$

which acts as the first-order time derivative of the metric on the hypersurface. We may now treat the triple (Σ, h_{ab}, K_{ab}) as a set of initial data for the second-order Einstein equation but first we will define more structure on Σ. Naturally we define the covariant derivative on the hypersurface as

$$D_e T^{ab...}{}_{cd....} = h^{e'}{}_e h^a{}_{a'} h^b{}_{b'} ... h^{c'}{}_c h^{d'}{}_d ... \nabla_{e'} T^{a'b'...}{}_{c'd'....} \tag{2.4}$$

which can be shown to obey

$$D_a h_{bc} = 0 \tag{2.5}$$

and is hence metric compatible. As D also inherits the torsion-free nature of ∇, the Fundamental Theorem of Riemannian Geometry guarantees that it is the Levi-Civita connection associated with h_{ab}. We are now also able to compute the three-dimensional Riemann tensor ${}^{(3)}R_{abcd}$ defined by the commutator of covariant derivatives

$$D_a D_b V_c - D_b D_a V_c = {}^{(3)}R_{abc}{}^d V_d \; . \tag{2.6}$$

This can be expanded in terms of the four-dimensional Riemann tensor R_{abcd} by Gauss' equation

$$ {}^{(3)}R_{abcd} = h^{a'}_{\ a} h^{b'}_{\ b} h^{c'}_{\ c} h^{d'}_{\ d} R_{a'b'c'd'} - K_{ac}K_{bd} + K_{ad}K_{bc} \tag{2.7} $$

which can be shown to preserve the expected symmetries of the Riemann tensor. If we start contracting components we arrive at the formulas for the Ricci tensor

$$ {}^{(3)}R_{ac} = h^{a'}_{\ a} h^{b'd'} h^{c'}_{\ c} R_{a'b'c'd'} - KK_{ac} + K_{ad}K_c^{\ d} \tag{2.8} $$

and the Ricci scalar

$$ {}^{(3)}R = R + 2R_{ac}n^a n^c - K^2 + K_{ab}K^{ab} \tag{2.9} $$

where R_{ab} and R are the four-dimensional Ricci tensor and Ricci scalar and $K = K_{ab}h^{ab} = \nabla_a n^a$ is the trace of the intrinsic curvature. Another useful relation is Codazzi's equation

$$ D_a K^a_{\ b} - D_b K = R_{ac} n^c h^a_{\ b} \tag{2.10} $$

which can be shown by computing $D_{[a}K_{b]c}$ and contracting with h^{ac}. The Gauss-Codazzi equations now encode the decomposition of the Ricci tensor into tangential and normal components to the hypersurface.

2.2 Foliation

Having studied one spacelike hypersurface, we will now layer space with many. Consider again the function $f(x^a)$ but now with the requirement that $f =$ const. for any value defines a unique spacelike hypersurface. This allows us to define a time-function t and label the spacelike hypersurfaces of constant time $\Sigma(t)$. To make sure that it is possible to choose such a time function, we must additionally assume that there are no obstructions such as changes in topology of the spacetime or boundaries that prevent us from choosing a global time function. We will describe later the mathematical requirement of the manifold to ensure this.
Let's also define a set of coordinates $\{x^i\}$ on the spacelike slices $\Sigma(t)$ such that the metric decomposes into the form

$$ ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt) \tag{2.11} $$

where α is the lapse and β^i is the shift. The spacelike metric γ_{ij} is equivalent to the induced metric h_{ab} on each slice $\Sigma(t)$ in the coordinate basis defined by $\{x^i\}$. Writing in matrix form, we hence find

$$ g_{ab} = \begin{pmatrix} -\alpha^2 + \beta_k\beta^k & \beta_i \\ \beta_j & \gamma_{ij} \end{pmatrix}, \ g^{ab} = \begin{pmatrix} -\frac{1}{\alpha^2} & \frac{\beta^i}{\alpha^2} \\ \frac{\beta^i}{\alpha^2} & \gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2} \end{pmatrix} \tag{2.12} $$

where we may assume $\alpha > 0$ with out loss of generality. In the coordinate chart $\{t, x^i\}$, the timelike normal is now $n_a = \lambda(1, 0, 0, 0)$ with $n^a = \lambda(-1/\alpha^2, \beta^i/\alpha^2)$. To ensure n_a is a unit vector, we hence pick $\lambda = -\alpha$.

2.3 Penrose Diagrams

In the above I promised that the triple (Σ, h_{ab}, K_{ab}) is a set of initial data for the second-order Einstein equation. We will now find out the limits of this by studying the causal structure of spacetime. The best way to do this is by visualising the different regions and asymptotic behaviours of any given spacetime. As spacetimes are (usually) infinitely big and four-dimensional, this is not an easy task and we would ideally like to make them more compact to look at a two-dimensional and finitely-sized diagram. This can be done through a process known as conformal compactification which preserves the causal structure of the spacetime and gives us a Penrose diagram. We will explore how to do this for Minkowski space but the procedure is similar for any other reasonable spacetime. The key insight to conformal compactification is that two spacetimes $(\mathcal{M}, g)$ and $(\mathcal{M}', g')$ with $\mathcal{M} \in \mathcal{M}'$ possess the same null geodesics if the metrics are related by a conformal transformation

$$g' = \omega^2 g \tag{2.13}$$

where ω is a positive function on the manifold. We now start with Minkowski spacetime in polar coordinates

$$ds^2 = -dt^2 + dr^2 + r^2 d\Omega^2 \tag{2.14}$$

where $d\Omega^2$ gives the line element on the angular space S^2. We notice the ranges of the coordinates are $-\infty < t < \infty$ and $0 \leq r < \infty$. Next we define advanced $u = t + r$ and retarded $v = t - r$ null coordinates so the metric takes the form

$$ds^2 = -du dv + \frac{1}{4}(v-u)^2 d\Omega^2 \tag{2.15}$$

where the new ranges of the coordinates are $-\infty < u \leq v < \infty$. We now choose compact coordinates $u = \tan p$ and $v = \tan q$ to give

$$ds^2 = \sec^2 p \sec^2 q \left(-dp dq + \frac{1}{4}\sin^2(p-q) d\Omega^2 \right) \tag{2.16}$$

with ranges $-\frac{\pi}{2} \leq p \leq q \leq \frac{\pi}{2}$. At this point the metric (2.16) is still the Minkowski metric but now we suffer a coordinate singularity where the metric degenerates at the edges of the coordinate ranges. This is expected to happen as we are cramming an infinite amount of real space into a finite coordinate range. To mitigate this issue, we perform a conformal transformation to get rid of the diverging factor

$$ds^2 = -dp dq + \frac{1}{4}\sin^2(p-q) d\Omega^2 \ . \tag{2.17}$$

We are now able to yet again change coordinates to $p = T - R$ and $q = T + R$ for a last time to see that our metric has turned into

$$ds^2 = -dT^2 + dR^2 + \frac{1}{4}\sin^2(2R) d\Omega^2 \tag{2.18}$$

the Einstein static universe with topology $\mathbb{R} \times S^3$ but importantly finite ranges on all the coordinates.

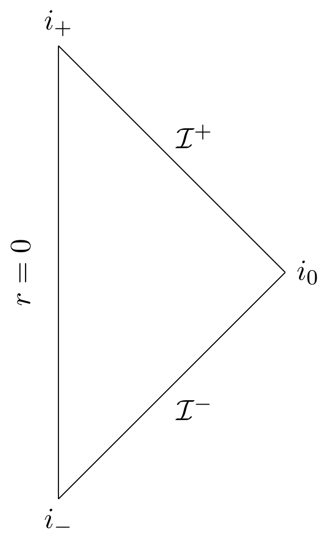

Figure 1: Penrose diagram of 4d Minkowski spacetime.

We may ignore the angular directions, as they do not add to the causal structure, and display the spacetime on a Penrose diagram where T forms the vertical axis and R the horizontal axis. This can be seen in fig. 1. It now looks like the spacetime has three boundaries. The line $R = r = 0$ is not a real boundary but rather the centre of the spacetime. As we have chosen to disregard angular directions, we have to remember that each point in the diagram, apart from the axis $r = 0$, corresponds to a transverse S^2 space with radius r. The other lines labelled with $\mathcal{I}^+$ and $\mathcal{I}^-$ correspond to future and past null infinities and are hence real boundaries of the spacetime (even if they are infinitely far away in real space!). The points labelled with i_- and i_+ are the past and future timelike infinities where all timelike curves originate and end up. Finally i_0 corresponds to spacelike infinity where all spacelike curves start and end.

The unphysical spacetime described by the metric (2.18) has the same null geodesics as the Minkowski spacetime we started in and hence future-directed light rays still travel upwards on 45° angles, originating at $\mathcal{I}^-$, reflecting at the centre $r = 0$ and ending on $\mathcal{I}^+$. Therefore light cones in the Penrose diagram look the same as we would expect in a regular space-time diagram of flat spacetime.

We are now equipped, in principle, to construct the Penrose diagram for any spacetime although in practise this may be complicated. Penrose diagrams in general allow us to look at purely the causal nature of the spacetime with all other complications stripped away.

2.4 Initial Value Problem

Any good classical theory has a well-posed initial value problem. This means that, given a valid set of initial data, we should be able to use the equations of motion to find a unique solution in a certain region of the spacetime. For general relativity, a theorem due to Madame Choquet-Bruhat guarantees that this holds true. We will now use causal structure to find out what region of spacetime we can find a solution for using a set of initial data. We choose a real scalar field ϕ on a general background with equation of motion

$$(\Box - m^2)\phi = 0 \tag{2.19}$$

where $\Box$ is the D'Alembert operator on our spacetime. If we include sources, this equation can be solved using a retarded Green's function defined by

$$(\Box - m^2)G(x,t;x',t') = \delta(x-x')\delta(t-t') \tag{2.20}$$

where (t',x') is the location of the delta function source. We write the field as

$$\begin{aligned}\phi(x,t) &= \int dt'\, d^3x'\ \delta(x-x')\delta(t-t')\phi(x',t') \\ &= \int dt'\, d^3x'\ (\Box - m^2)G(x,t;x',t')\phi(x',t')\ .\end{aligned} \tag{2.21}$$

Now we note the retarded Green's function $G(x,t;x',t')$ vanishes outside of the past light cone starting on (t,x). It is therefore possible to choose an integration volume just outside of the past light cone without loss of generality.

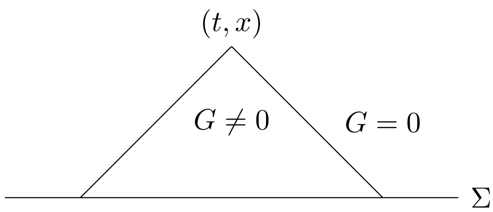

Figure 2: Past light cone starting on (t,x).

This is shown in fig. 2 where we choose integration just outside the past light cone and along the spacelike hypersurface Σ. Integrating by parts, we get a bulk term that vanishes by the Klein-Gordon equation (2.19) and a boundary term that vanishes everywhere except on Σ. This boundary term is

$$\phi(x,t) = \int_\Sigma d^3x'\ \phi\dot{G} - \dot{\phi}G \tag{2.22}$$

where we can see that we need only specify ϕ and π on the portion of Σ intersecting the past light cone to find the full solution.

This motivates us to define the inverse of what we have just constructed. The future domain of dependence $D^+(\Sigma)$ of a spacelike hypersurface Σ is the set of points $p \in \mathcal{M}$ such that every past-inextendible causal curve starting on p has to intersect Σ. The past domain of dependence $D^-(\Sigma)$ is defined similarly as the set of points $p \in \mathcal{M}$ such that every future-inextendible curve starting on p has to intersect Σ. The union of these regions is known as the domain of dependence $D(\Sigma) = D^+(\Sigma) \cup D^-(\Sigma)$. It is easy to see now that specifying initial data (ϕ,π) on Σ allows us to use retarded and advanced Green's functions

to determine the full solution in $D(\Sigma)$. This results turns out to be true for any classical theory with a well-posed initial value problem.
If there exists a surface Σ such that $D(\Sigma) = \mathcal{M}$ then the manifold $(\mathcal{M}, g)$ is said to be globally hyperbolic. In this case there exists a theorem that guarantees we can choose a global time function, as discussed above. The surface Σ is then called a Cauchy hypersurface.

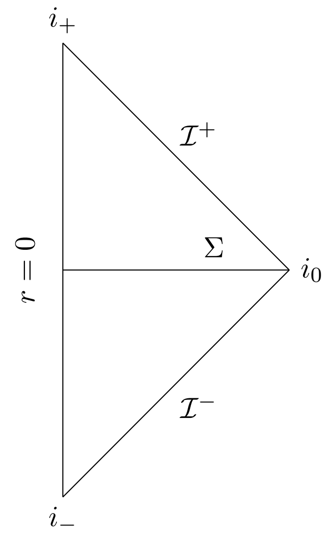

Figure 3: Cauchy hypersurface in Minkowski spacetime.

For Minkowski spacetime, we can use the Penrose diagram as shown in fig. 3 to easily see that the domain of dependence associated with the Cauchy hypersurface Σ covers the whole manifold. Hence flat spacetime is globally hyperbolic. Since t is one of our usual coordinates in Minkowski spacetime, we may choose this as global time function.

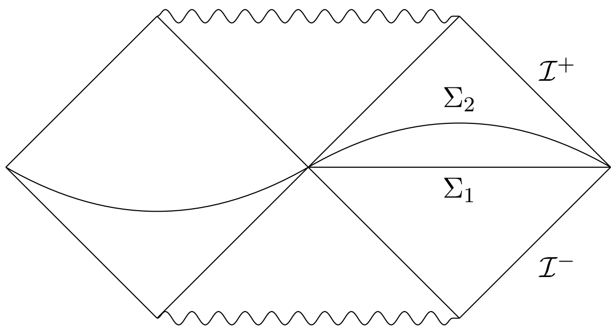

Figure 4: Hypersurfaces in maximally extended Schwarzschild spacetime.

For Schwarzschild spacetime the situation is not so simple anymore. We can see two different choices of hypersurface in fig. 4. Physically, we should really pick Σ_1 since we only have access to data outside the event horizon but then we run into the problem that only the exterior of the black hole is included in the domain of dependence so we cannot treat the interior of the black hole as physical if we want a globally hyperbolic spacetime. To fix this, we may choose the hypersurface Σ_2 in which case clearly the whole maximal analytic extension of the Schwarzschild metric is included in the domain of dependence. However, this is not very helpful in real initial value problems since we will never have access to the data of the parallel universe. Therefore we go back to Σ_1 and treat the interior of the black hole as unknown. Cutting out this region yields a globally hyperbolic spacetime with global time function t in Schwarzschild coordinates.

3 Hamiltonian Formulation of Gravity

We now combine all of the tricks we learnt so far and formulate a Hamiltonian theory of classical general relativity. We first choose a foliation $\Sigma(t)$ of spacetime and write the metric in the form

$$ds^2 = -\alpha^2 dt^2 + \gamma_{ij}(dx^i + \beta^i dt)(dx^j + \beta^j dt) \tag{3.1}$$

which has the properties discussed earlier. On the hypersurface we now find the extrinsic curvature

$$K_{ij} = h^a_{\ i} h^b_{\ j} \nabla_a n_b \tag{3.2}$$

where $h_{ab} = g_{ab} + n_a n_b$ is the induced metric on each $\Sigma(t)$ and n_a is the unit normal. Using $n_a = (-\alpha, 0, 0, 0)$ we find

$$h^0_{\ 0} = 0,\ h^i_{\ 0} = -\beta^i,\ h^0_{\ i} = 0,\ h^i_{\ j} = \delta^i_{\ j} \tag{3.3}$$

and so the extrinsic curvature simplifies to

$$K_{ij} = \Gamma^0_{ij} n_0 \tag{3.4}$$

where

$$\begin{aligned}\Gamma^0_{ij} &= \frac{1}{2} g^{00}(\partial_i g_{0j} + \partial_j g_{0i} - \partial_0 g_{ij}) + \frac{1}{2} g^{0k}(\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}) \\ &= \frac{1}{2}\left(-\frac{1}{\alpha^2}\right)(-\dot{\gamma}_{ij} + \partial_i \beta_j + \partial_j \beta_i) + \frac{\beta_k}{\alpha^2}\,{}^{(3)}\Gamma^k_{ij} = -\frac{1}{2\alpha}(D_i\beta_j + D_j\beta_i - \dot{\gamma}_{ij})\,.\end{aligned} \tag{3.5}$$

Gauss' and Codazzi's equations together with $\sqrt{-g} = \alpha\sqrt{\gamma}$ allow us to decompose the Einstein-Hilbert action as

$$\begin{aligned}S &= \frac{1}{16\pi}\int d^4x \sqrt{-g}\,(R - 2\Lambda) \\ &= \frac{1}{16\pi}\int dt\, d^3x\, \alpha\sqrt{\gamma}({}^{(3)}R + K^{ij}K_{ij} - K^2 + 2R_{ab}n^a n^b - 2\Lambda)\end{aligned} \tag{3.6}$$

where $2R_{ab}n^a n^b$ is a boundary term. From now on, we will drop the pre-factor of the action since it does not affect the equations of motion. As an aside, the Dirac delta function $\delta(x - x')$ looks like a scalar but is actually a density of weight -1. This usually does not make a difference but when applying covariant derivatives

$$\nabla_j \delta(x - x') = \partial_j \delta(x - x') - \Gamma^k_{kj}\delta(x - x') \tag{3.7}$$

we see that it behaves differently. This behaviour is important to make sure that the definition of the delta function

$$f(x) = \int d^3x' \sqrt{\gamma}\, \delta(x - x') f(x') \tag{3.8}$$

is consistent in curved spacetime. We now want to construct a Hamiltonian of gravity. The canonical momenta are derived from the action as

$$\pi = 0,\ \pi^i = 0,\ \pi^{ij} = \sqrt{\gamma}(K^{ij} - \gamma^{ij}K) \tag{3.9}$$

where π is associated with α, π^i is associated with β^i and π^{ij} is the canonical momentum with respect to γ^{ij}. We notice that many of the momenta are zero but since this also happened in electromagnetism, we now know how to deal with this. Our first attempt at a Hamiltonian is then

$$\begin{aligned} H &= \int d^3x\ \pi^{ij}\dot{\gamma}_{ij} - \alpha\sqrt{\gamma}({}^{(3)}R + K^{ij}K_{ij} - K^2 - 2\Lambda) + \nu\pi + \lambda_i\pi^i \\ &= \int d^3x\ \frac{\alpha}{\sqrt{\gamma}}\pi^{ij}\pi_{ij} - \frac{1}{2}\frac{\alpha}{\sqrt{\gamma}}(\pi^k_{\ k})^2 - \alpha\sqrt{\gamma}({}^{(3)}R - 2\Lambda) - 2\beta_i D_i\pi^{ij} + \nu\pi + \lambda_i\pi^i \end{aligned} \tag{3.10}$$

where we have imposed the zero momenta as primary constraints with Lagrange multipliers ν and λ_i. We can again find secondary constraints by finding the time derivatives of the constraints

$$\dot{\pi} = \frac{\delta H}{\delta\alpha} = \frac{1}{\sqrt{\gamma}}\left(\pi^{ij}\pi_{ij} - \frac{1}{2}(\pi^k_{\ k})^2\right) - \sqrt{\gamma}({}^{(3)}R - 2\Lambda) = 0 \tag{3.11}$$

known as the Hamiltonian constraint $\mathcal{H}$ and

$$\dot{\pi}^i = \frac{\delta H}{\delta\beta^i} = -D_j\pi^{ij} = 0 \tag{3.12}$$

known as the diffeomorphism constraint χ^i. The correct Hamiltonian is therefore

$$H = \int d^3x\ \pi\dot{\alpha} + \pi^i\beta_i + \alpha\mathcal{H} + \beta_i\chi^i \tag{3.13}$$

where we have determined the Lagrange multipliers. The Hamiltonian consists only of constraints and vanishes. This is not a problem however as we only care about the variation of the Hamiltonian which does not vanish in general. To obtain the evolution equations, we must pick a gauge. We pick Gaussian gauge $\alpha = \beta^i = 0$ and can find the equations of motion using Poisson brackets. Before we do this, we make an observation about the Hamiltonian constraint. The Hamiltonian constraint can be written in the suggestive form

$$\mathcal{H} = G_{ijkl}\pi^{ij}\pi^{kl} - \sqrt{\gamma}({}^{(3)}R - 2\Lambda) \tag{3.14}$$

which vaguely looks like a Hamiltonian for a particle moving in a potential. We will get back to this later. The brackets mostly become trivial at this point except

$$\dot{\gamma}_{ij} = 2G_{ijkl}\pi^{kl} \tag{3.15}$$

and $\dot{\pi}_{ij}$ which together reproduce the Einstein equation. Therefore given initial data $(\Sigma, \gamma_{ij}, \pi_{ij})$ satisfying the Hamiltonian and diffeomorphism constraints, we may use the equations of motion to evolve it forward and backward in time within the domain of dependence $D(\Sigma)$.

The Hamiltonian is also modified if we add matter to the equation. For example consider a scalar field with matter action

$$S = \int d^4x\sqrt{-g}\ -\frac{1}{2}g^{ab}\partial_a\phi\partial_b\phi - V(\phi) \tag{3.16}$$

where $V(\phi)$ is a general polynomial potential term. Expanding the foliation metric we can rewrite the action as

$$S = \int dt\, d^3x\alpha\sqrt{\gamma}\ \left(\frac{1}{2}\frac{\dot{\phi}^2}{\alpha^2} - \frac{\beta^i}{\alpha^2}\dot{\phi}\partial_i\phi - \frac{1}{2}\left[\gamma^{ij} - \frac{\beta^i\beta^j}{\alpha^2}\right]\partial_i\phi\partial_j\phi - V(\phi)\right) \tag{3.17}$$

where we may read off the canonical momentum

$$\pi_\phi = \sqrt{\gamma}\frac{\dot{\phi}}{\alpha} - \sqrt{\gamma}\frac{\beta^i}{\alpha}\partial_i\phi\ . \tag{3.18}$$

Adding a scalar field hence adds the term

$$\int d^3x\ \alpha\left(\frac{\pi_\phi^2}{2\sqrt{\gamma}} + \frac{1}{2}\sqrt{\gamma}\gamma^{ij}\partial_i\phi\partial_j\phi + \sqrt{\gamma}V(\phi)\right) + \beta^i\pi_\phi\partial_i\phi \tag{3.19}$$

to the Hamiltonian. The first term modifies the Hamiltonian constraint to $\mathcal{H}+\frac{1}{2}$(matter energy density) and the second term modifies the diffeomorphism constraint to $\chi_i + \pi_\phi\partial_i\phi$. We can now use these modified constraints to follow the same procedure for finding the equations of motion as for vacuum gravity.

3.1 Hamiltonian Constraint

To understand the evolution of the metric in the canonical approach, we study the Hamiltonian constraint

$$\mathcal{H} = G_{ijkl}\pi^{ij}\pi^{kl} - \sqrt{\gamma}({}^{(3)}R - 2\Lambda) \tag{3.20}$$

where we now explicitly identify the terms with the kinetic and potential terms of a particle. Writing down the action for a particle in a potential

$$S = \int ds\ \frac{1}{2}g_{ab}\dot{x}^a\dot{x}^b - V(x) \tag{3.21}$$

we can compute the canonical momenta $p_a = g_{ab}\dot{x}^b$ and find the Hamiltonian

$$H = \frac{1}{2}g^{ab}p_ap_b + V(x) \tag{3.22}$$

where we learn that the factor G_{ijkl} multiplying the momenta behaves like an inverse metric. The metric G^{ijkl} this gives should have the property

$$G^{ijkl}G_{klmn} = \frac{1}{2}(\delta^i{}_m\delta^j{}_n + \delta^j{}_m\delta^i{}_n) \tag{3.23}$$

to preserve the symmetries of the tensors. We can therefore construct the metric

$$G^{ijkl} = \frac{1}{2}\sqrt{\gamma}(\gamma^{ik}\gamma^{jl} + \gamma^{il}\gamma^{jk} - 2\gamma^{ij}\gamma^{kl}) \tag{3.24}$$

which gives a measure on the space of symmetric rank 2 tensors, superspace, on the space-like hypersurface $\Sigma(t)$, parametrised by γ_{ij}. Suppose we have two metrics γ_{ij} and $\gamma_{ij}+d\gamma_{ij}$. We are now able to compute their distance in superspace as

$$ds^2 = G^{ijkl}d\gamma_{ij}d\gamma_{kl} \tag{3.25}$$

where we say that any pair of 3-geometries separated by $d\gamma_{ij}$ are apart a distance of ds at a point. To find the distance between two geometries in the whole space, we must integrate

$$dS^2 = \int d^3x \; G^{ijkl}d\gamma_{ij}d\gamma_{kl} \tag{3.26}$$

which defines the DeWitt metric. This metric has the nice property that two metrics separated by a diffeomorphism $d\gamma_{ij} = D_{(i}\xi_{j)}$ give $dS = 0$ and so the measure correctly picks out physically equivalent configurations.
For any metric perturbation $d\gamma_{ij}$, we have a 3x3 symmetric matrix and so six components. Therefore we should be able to represent G^{ijkl} in a more usual form as a 6x6 matrix. Take for example a conformal transformation where $d\gamma_{ij} = \lambda\gamma_{ij}$. The distance function can be written as

$$G^{ijkl}d\gamma_{ij}d\gamma_{kl} = -6\sqrt{\gamma}\lambda^2 \tag{3.27}$$

where we see that the 6x6 metric cannot be Riemannian as the distance has a negative sign. Hence we pick coordinates $\zeta = \sqrt{\frac{32}{3}}\gamma^{1/4}$ and $\zeta^A,\ A = 1,2,3,4,5$ on the space of symmetric tensors where ζ parametrises conformal transformations and ζ^A describe other transformations orthogonal to this. The metric in these coordinates is

$$G_{AB} = \begin{pmatrix} -1 & 0 \\ 0 & \zeta^2\bar{G}_{AB} \end{pmatrix} \tag{3.28}$$

which is a cone over $\bar{G}_{AB}$

$$ds^2 = -d\zeta^2 + \zeta^2\bar{G}_{AB}d\zeta^A d\zeta^B \; . \tag{3.29}$$

The geometry $\bar{G}_{AB}$ can also be written as

$$\bar{G}_{AB} = \mathrm{tr}(\gamma^{-1}\gamma_{,A}\gamma^{-1}\gamma_{,B}) \tag{3.30}$$

where we trace over the spatial indices on the hypersurface $\Sigma(t)$. The curvature of $\bar{G}_{AB}$ is

$$R_{ABCD} = k(\bar{G}_{AC}\bar{G}_{BD} - \bar{G}_{AD}\bar{G}_{BC}),\ k < 0 \tag{3.31}$$

where we can see that $\bar{G}_{AB}$ is positive-definite and so must be a 5d hyperboloid. We also see that R is covariantly constant and hence the 5d space must be maximally symmetric.

This strongly restricts the options for possible spaces. In fact the space of 3x3 symmetric matrices with unit determinant (as we fixed scaling) is $SL(3)/SO(3)$ which matches our dimension of 5. Now the overall scalar curvature including scaling is

$$ {}^{(6)}R = -\frac{60}{\zeta^2} \tag{3.32} $$

which tells us there is some form of singularity as $\zeta \to 0$. Topologically, we see that G_{AB} lives on $SL(3)/SO(3) \oplus \mathbb{R}^+$ at each point in space.

3.2 Hamiltonian Einstein Equation

We would now like to analyse the Hamiltonian Einstein equation and see if we can extract any information about its solutions. The Einstein equation derived from action principles is equivalently reproduced by

$$ \dot{\gamma}_{ij} = \{\gamma_{ij}, H\}, \ \dot{\pi}_{ij} = \{\pi_{ij}, H\} \ . \tag{3.33} $$

To solve these, we pick a convenient gauge

$$ \alpha = \frac{1}{2} \frac{1}{\sqrt{\int d^3x \sqrt{\gamma} \ {}^{(3)}R}}, \ \beta^i = 0 \tag{3.34} $$

such that the equations are equivalent to the system of second order ODEs

$$ \frac{d^2\gamma_{ij}}{dt^2} - \frac{d\gamma_{ik}}{dt}\frac{d\gamma_{jl}}{dt}\gamma^{kl} + \frac{1}{2}\frac{d\gamma_{ij}}{dt}\frac{d\gamma_{kl}}{dt}\gamma^{kl} + \frac{1}{4\sqrt{\gamma}}\gamma_{ij}G^{klmn}\frac{d\gamma_{kl}}{dt}\frac{d\gamma_{mn}}{dt} = -2\left({}^{(3)}R_{ij} - \frac{1}{4}\gamma_{ij}\ {}^{(3)}R\right) \ . \tag{3.35} $$

This is remarkable because without the RHS curvature term, we exactly have the geodesic equation in $SL(3)/SO(3) \oplus \mathbb{R}^+$ at each point in space. The curvature term is responsible for coupling together different points in space like a force. Fixing α and β^i, the spacetime traces out a curve in the space of all metrics on surfaces of constant time $\Sigma(t)$. We will call this space of 3-metrics modulo coordinate transformations superspace. Hence any spacetime is a trajectory in superspace.

So far we have ignored the possibility for boundary terms however if we are dealing with an asymptotically flat, as seen above, or an asymptotically Anti-deSitter (AdS) spacetime, we may add an extra piece

$$ \frac{1}{16\pi}\int dS^k \alpha\sqrt{\gamma}\gamma^{ij}(\gamma_{ik,j} - \gamma_{ij,k}) \tag{3.36} $$

to the Hamiltonian. For asymptotically flat or asymptotically AdS spacetimes, this is the ADM mass.

3.2.1 Cosmology

Restricting the number of degrees of spacetime even further, we can simlify our motion in superspace to minisuperspace. This is the case for a simple cosmological model of the FLRW metric in a topologically closed universe

$$ ds^2 = -dt^2 + a(t)^2 d\Omega_3^2 \tag{3.37} $$

where $d\Omega_3^2$ is the round metric on spatial S^3 slices. We therefore identify $\gamma_{ij}dx^i dx^j = a(t)^2 d\Omega_3^2$ where our only degree of freedom is the scale factor $a(t)$. We see that the metric is already in a gauge $\alpha = 1$ and $\beta^i = 0$ so we can find the Ricci curvature

$$ {}^{(3)}R_{ij} = \frac{2\gamma_{ij}}{a^2}, \; {}^{(3)}R = \frac{6}{a^2} \tag{3.38} $$

and the extrinsic curvature

$$ K_{ij} = a\dot{a}\gamma_{ij} \; . \tag{3.39} $$

This allows us to write the action in minisuperspace as

$$ S = \int dt\, d^3x\, \sqrt{\gamma}({}^{(3)}R + K_{ij}K^{ij} - K^2 - 2\Lambda) = \int dt\; 6a - 6a\dot{a}^2 - 2\Lambda a^3 \tag{3.40} $$

where we have dropped an overall numerical factor in the last step. The canonical momentum for a is $\pi_a = -12a\dot{a}$. This gives the Hamiltonian

$$ H = -\frac{\pi_a^2}{24a} - 6a + 2\Lambda a^3 \tag{3.41} $$

where we again see that the kinetic term for a has the wrong sign because it corresponds to a conformal transformation. Hamilton's equations become

$$ \dot{a} = -\frac{\pi_a}{12a}, \; \dot{\pi_a} = 6 - 6\Lambda a^2 - \frac{\pi_a^2}{24a^2} \; . \tag{3.42} $$

This can be solved for $\Lambda > 0$ to get the well-known de Sitter metric

$$ a(t) = \sqrt{\frac{\Lambda}{3}} \cosh\left(\sqrt{\frac{3}{\Lambda}}t\right), \; \pi_a(t) = -\sqrt{12\Lambda}\sinh\left(2\sqrt{\frac{3}{\Lambda}}t\right) \tag{3.43} $$

of an exponentially expanding universe. It turns out for $\Lambda \leq 0$, there are no real solutions to (3.42).

3.2.2 Kasner Universe

Another interesting geometry to look at in minisuperspace is the Kasner universe. It is believed to describe the behaviour of spacetime close to a singularity like the big bang. In Cartesian coordinates, we write

$$ ds^2 = -dt^2 + a(t)^2dx^2 + b(t)^2dy^2 + c(t)^2dz^2 \tag{3.44} $$

which is a generally anisotropic but homogeneous version of the FLRW metric. On surfaces of constant t, the extrinsic curvature is

$$ K_{ij} = \mathrm{diag}(a\dot{a}, b\dot{b}, c\dot{c}) \tag{3.45} $$

and ${}^{(3)}R = 0$. The minisuperspace action is hence

$$ S = \int dt\; abc\left(-\frac{2\dot{a}\dot{b}}{ab} - \frac{2\dot{b}\dot{c}}{bc} - \frac{2\dot{c}\dot{a}}{ca}\right) \tag{3.46} $$

which gives canonical momenta $\pi_a = -2(bc)\dot{}$, $\pi_b = -2(ca)\dot{}$, $\pi_c = -2(ab)\dot{}$. We again determine the Hamiltonian

$$H = \frac{1}{8}\left(\frac{a\pi_a^2}{bc} + \frac{b\pi_b^2}{ca} + \frac{c\pi_c^2}{ab}\right) - \frac{1}{4}\left(\frac{\pi_a\pi_b}{c} + \frac{\pi_b\pi_c}{a} + \frac{\pi_c\pi_a}{b}\right) \tag{3.47}$$

and find Hamilton's equations

$$\dot{a} = \frac{1}{4bc}(a\pi_a - b\pi_b - c\pi_c),\ \dot{\pi}_a = -\frac{\pi_a^2}{bc} + \frac{1}{8a^2bc}(\pi_b b - \pi_c c)^2 \tag{3.48}$$

with the remaining equations differing by a cyclic permutation of a, b and c. These equations can be solved using an ansatz $a = t^{p_1}$, $b = t^{p_2}$, $c = t^{p_3}$ and enforcing $H = 0$. These give the conditions $p_1p_2 + p_2p_3 + p_3p_1 = 0$ and $p_1 + p_2 + p_3 = 1$. Combining this with the Hamiltonian constraint we obtain $p_1^2 + p_2^2 + p_3^2 = 1$ and $p_1 + p_2 + p_3 = 1$. This means we can either have two p_i positive and one negative or have two p_i zero and the last equal to one. The latter gives Rindler spacetime which can be written as

$$ds^2 = -dt^2 + t^2dx^2 + dy^2 + dz^2\ . \tag{3.49}$$

As an aside, for any choice of p_i, the conditions ensure that $abc \sim t$ and hence the volume of the spacetime is linearly increasing in time. We also have $R \sim 1/t^2$ and so we always have a singularity at $t = 0$. This is necessary if we want the metric to describe the spacetime near the big bang. We will study the dynamics of this spacetime in more detail later.

4 Canonical Quantisation of Gravity

One of the most interesting features of having a Hamiltonian formulation of gravity is that it is now possible to embed gravity into the framework of quantum mechanics using canonical quantisation. Thereby we would like to promote the observables to Hermitian operators and replace Poisson or Dirac brackets

$$\{\gamma_{ij}(x), \pi^{kl}(x')\} = \frac{1}{2}(\delta^k_i\delta^l_{\ j} + \delta^k_j\delta^l_{\ i})\delta(x - x') \tag{4.1}$$

by commutators. One representation of this algebra maps

$$\pi^{ij}(x) \to -i\hbar\frac{\delta}{\delta\gamma_{ij}(x)} \ . \tag{4.2}$$

which allows us to promote the constraints $\mathcal{H}$ and χ_i to operators $\hat{\mathcal{H}}$ and $\hat{\chi}_i$. As classically $\mathcal{H} = \chi_i = 0$, we need to impose

$$\hat{\mathcal{H}}\psi = \chi_i\psi = 0 \tag{4.3}$$

on all states ψ in the Hilbert space.

4.1 Wheeler-DeWitt Equation

We will now look at the constraint equation $\hat{\mathcal{H}}\psi = 0$ in more detail. This equation is known as the Wheeler-DeWitt equation and is key to producing a consistent quantum theory of gravity. We can expand this as

$$\left[-\hbar^2 G_{ijkl}\frac{\delta}{\delta\gamma_{ij}}\frac{\delta}{\delta\gamma_{kl}} - \sqrt{\gamma}\left(R - 2\Lambda + \frac{1}{2}(\text{matter energy density})\right)\right]\psi = 0 \tag{4.4}$$

where the first term again acts like a massless Klein-Gordon equation at each point in space and the second term provides an effective mass. Naturally, this equation inherits the same interpretational problem as Klein-Gordon wave functions (for example negative probabilities) but we will ignore this for now and push on with the Klein-Gordon analysis. Consider a Klein-Gordon equation

$$(-\hbar^2\Box + m^2)\phi = 0 \tag{4.5}$$

in any spacetime and use a semiclassical WKB approximation $\phi = Ae^{iS/\hbar}$ to solve it. Expanding in powers of $\hbar$, we get

$$\Box\phi = g^{ab}\left(\nabla_a\nabla_b A + \frac{2i}{\hbar}\nabla_a S\nabla_b A + \frac{i}{\hbar}A\nabla_a\nabla_b S - \frac{1}{\hbar^2}A\nabla_a\nabla_b S\right)e^{\frac{i}{\hbar}S} \tag{4.6}$$

where we can see that if we take $\hbar \to 0$ in the classical limit, the Klein-Gordon equation turns into

$$\nabla_a S\nabla^a S + m^2 = 0 \ . \tag{4.7}$$

We can then add a little bit of quantumness at the next order

$$2\nabla_a S \nabla^a A + iA\nabla_a \nabla^a S = 0 \tag{4.8}$$

which modifies the amplitude A of the wave function. Identifying $\nabla_a S$ with the momentum p_a reproduces the expected dispersion relation $p^2 + m^2 = 0$. We can also show that this momentum p_a objeys the geodesic equation

$$p^a \nabla_a p_b = \nabla^a S \nabla_a \nabla_b S = \nabla^a \nabla_b \nabla_a S = \frac{1}{2}\nabla_b(\nabla_a S \nabla^a S) = \frac{1}{2}\nabla_b(-m^2) = 0\ . \tag{4.9}$$

Now we would like to import the WKB approximation for the Wheeler-DeWitt equation to see how quantum gravity reproduces classical gravity in this limit. We will have a wave function ψ(3-geometry) that encodes the quantum mechanical distribution in the space of 3-geometries. The WKB approximation should then allow us to reproduce the classical behaviour with quantum corrections.

4.1.1 Path Integral Formulation

Having quantised by turning everything into operators, we take a quick detour to explore what happens when we fully get rid of operators. Of course this must lead to the same physics. Consider the path integral

$$Z = \int D[g]\ e^{\frac{i}{\hbar}S[g]} \tag{4.10}$$

over all 4-metrics with fixed boundary on a spacelike hypersurface Σ. Varying the metric $g_{ab} \to g_{ab} + h_{ab}$ in the Einstein-Hilbert action

$$\delta S = \int d^4x\sqrt{-g}\ \left[\left(-R_{ab} + \frac{1}{2}Rg_{ab}\right)h^{ab} - \Box h_{ab} + \nabla_a \nabla_b h^{ab}\right] \tag{4.11}$$

gives rise to a bulk term that produces the Einstein equation and a boundary term. We may use the divergence theorem on the boundary term to obtain

$$\delta S = \int d\Sigma^a\ \ - \nabla_a h + \nabla_b h^b{}_a \tag{4.12}$$

where we are integrating over the boundary. The integrand is sometimes referred to as a presymplectic potential. Now consider the boundary Σ with a normal n_a and an induced metric γ_{ab}. The metric perturbation h_{ab} can be decomposed

$$h_{ab} = An_a n_b - (B_a n_b + n_a B_b) + C_{ab} \tag{4.13}$$

into components normal, normal-tangent and tangent to the hypersurface Σ. This allows us to write the boundary term as

$$\delta S = \int d^3x\sqrt{\gamma}\ [-n^a\partial_a C + AK - 2n^a B_b + D_a B^a - C^{ab}K_{ab}]\ . \tag{4.14}$$

To obtain a well-posed variational principle we must hence add the GHY boundary term

$$\delta S_{GHY} = 2 \int d^3x \sqrt{\gamma}\, K \tag{4.15}$$

where $K = \nabla_a n^a$ is the trace of the extrinsic curvature. To preserve $n_a n^a = -1$, we take $\delta n_a = 1/2 A n_a$ and $\delta n^a = \delta n_a g^{ab} - h^{ab} n_b = B^a - 1/2 A n^a$. Now the GHY boundary term is

$$\delta S_{GHY} = \int d^3x \sqrt{\gamma}\, [-2 D_a B^a + 2 a_a B^a - AK + CK + n^a \partial_a C] \tag{4.16}$$

where a_a is the acceleration. Evaluating both boundary terms across the boundary gives

$$\delta S + \delta S_{GHY} = \int d^3x \sqrt{\gamma}\, \left[-D_a B^a - C^{ab} \left(K_{ab} - \frac{1}{2} K \gamma_{ab} \right) \right] \tag{4.17}$$

where the first term vanishes as the boundary of the boundary is empty. The second term is gives only components lying in the surface and defines the Brown-York stress tensor $K_{ab} - 1/2 K g_{ab}$. Inserting the Einstein-Hilbert action with the GHY boundary term back into the path integral,

$$Z[\gamma] = \int D[g]\; e^{\frac{i}{\hbar}(S + S_{GHY})} \tag{4.18}$$

we find that now $Z[\gamma]$ obeys the Wheeler-DeWitt equation. The canonical momentum related to γ_{ij} is now given by the Brown-York stress tensor.

4.2 Quantising De Sitter Spacetime

Having explored the general approach to canonically quantising gravity, we now turn to an example to illustrate the insights we get from doing this. Consider a maximally symmetric spacetime with positive cosmological constant $\Lambda > 0$. As discussed above, the classical metric is given by

$$ds^2 = -dt^2 + a^2(t) d\Omega_3^2 \tag{4.19}$$

where $d\Omega_3^2$ is the line element on S^3 with round metric and $a(t) = \sqrt{3/\Lambda} \cosh(\sqrt{\Lambda/3} t)$. Due to its many symmetries, this spacetime only has one free parameter a that can be measured. Hence quantum mechanically we are interested in amplitudes

$$\langle a_f | a_i \rangle = \int_{a_i}^{a_f} D[g] e^{\frac{i}{\hbar} S[g]} \tag{4.20}$$

corresponding to transitions in the scale factor from a_i to a_f. The minisuperspace Hamiltonian is easily computed as

$$H = -\frac{\pi_a^2}{24a} - 6a + \Lambda a^2 \tag{4.21}$$

where we choose a representation $\pi_a = -i\hbar\,\partial/\partial a$ for the canonical momentum operator when quantising. The Wheeler-DeWitt equation $\hat{\mathcal{H}}\psi(a) = 0$ then becomes

$$\frac{\hbar^2}{24a}\frac{\partial^2\psi}{\partial a^2} + (-6a + 2\Lambda a^3)\psi = 0 \tag{4.22}$$

where $\psi(a)$ is now a wave function describing the quantum mechanical distribution of the value of the scale factor a. The form of the Wheeler-DeWitt equation reminds a lot of the Schrödinger equation in this case. The "wrong" sign of the kinetic term again stems from the fact that a plays the role of a dynamic conformal factor in the metric. As promised above we now import the WKB approximation

$$\psi = Ae^{\frac{i}{\hbar}S} \tag{4.23}$$

to solve the Wheeler-DeWitt equation. At lowest order $O(\hbar^{-2})$, the equation becomes

$$-\left(\frac{\partial S}{\partial a}\right)^2 - 144a^2 + 48\Lambda a^2 = 0 \tag{4.24}$$

and the $O(\hbar^{-1})$ correction to the amplitude is given by

$$\left(\frac{\partial^2 S}{\partial a^2}\right)\Lambda + 2\frac{\partial A}{\partial a}\frac{\partial S}{\partial a} = 0\ . \tag{4.25}$$

The $O(\hbar^{-2})$ equation is separable

$$\frac{\partial S}{\partial a} = \pm 12a\sqrt{\frac{\Lambda a^3}{3} - 1} \tag{4.26}$$

and exactly solvable

$$S = \pm\frac{12}{\Lambda}\left(\frac{\Lambda a^2}{3} - 1\right)^{\frac{3}{2}} \tag{4.27}$$

which returns the classical solution. At next order we find an equation for A

$$\frac{\partial}{\partial a}\ln A = -\frac{1}{2}\frac{\partial}{\partial a}\ln\frac{\partial S}{\partial a} \tag{4.28}$$

where we find

$$A = \left(\frac{\Lambda a^2}{3} - 1\right)^{-\frac{1}{4}} a^{-\frac{1}{2}}\ . \tag{4.29}$$

This gives us a general solution to the wave function in the WKB approximation

$$\psi = \left(\frac{\Lambda a^2}{3} - 1\right)^{-\frac{1}{4}} a^{-\frac{1}{2}}\left(\alpha e^{\frac{i}{\hbar}\left[\frac{12}{\Lambda}\left(\frac{\Lambda a^2}{3}-1\right)^{\frac{1}{2}}\right]} + \beta e^{-\frac{i}{\hbar}\left[\frac{12}{\Lambda}\left(\frac{\Lambda a^2}{3}-1\right)^{\frac{1}{2}}\right]}\right) \tag{4.30}$$

where α and β are constants of integration that may be chosen such that ψ is normalised over the range of a. To calculate the amplitude for transitioning from a_i to a_f, we remember that the classical solution dominates the path integral over large time periods with quantum corrections. Consider $a_i < a_f$, as we would expect for an expanding universe. Taking some initial time $t_0 > 0$, the classical equations of motion permit two solutions

$$a(t) = \sqrt{\frac{3}{\Lambda}} \cosh\left(\sqrt{\frac{3}{\Lambda}}(t - t_0)\right) \text{ or } a(t) = \sqrt{\frac{3}{\Lambda}} \cosh\left(\sqrt{\frac{3}{\Lambda}}(-t - t_0)\right) \tag{4.31}$$

which means that any given a_i at some time $t_i > t_0$ there are two classical paths to a_f at $t_f > t_i$. The path given by the first equation is shorter and corresponds to a monotone expanding universe. The longer path contracts and then expands again to reach a_f.

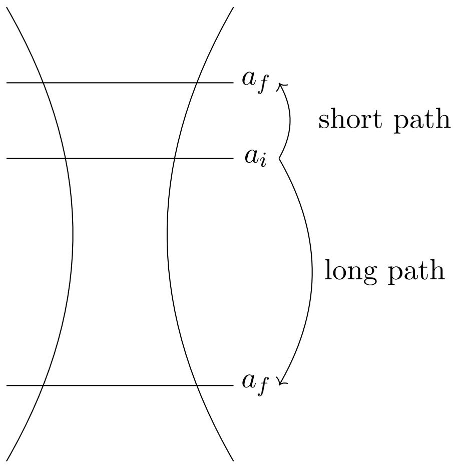

Figure 5: Classical solutions to a trajectory from a_i to a_f.

This can be seen in fig. 5 where the universe has a characteristic minimum scale factor $a_{min} = \sqrt{3/\Lambda}$. Returning to the quantum picture, we choose the short path and evaluate the action

$$S_{class} = \int_{t_i}^{t_f} dt\, (6a - 6a\dot{a}^2 - 2\Lambda a^3) = -\frac{12}{\Lambda}\left[\left(\frac{\Lambda a_f^2}{3} - 1\right)^{\frac{3}{2}} - \left(\frac{\Lambda a_i^2}{3} - 1\right)^{\frac{3}{2}}\right] \tag{4.32}$$

which we can insert into the path integral to find

$$\langle a_f | a_i \rangle \approx \mathcal{N} e^{\frac{i}{\hbar} S_{class}} \tag{4.33}$$

along the short path where $\mathcal{N}$ is an irrelevant numerical factor. We now see that the two terms in ψ correspond to the short, expanding path and the long, contracting path. As expected, the quantum mechanical system takes both permitted paths from a_i to a_f as once.

4.2.1 Causal Structure of De Sitter Spacetime

We will now take a brief detour to the causal structure of De Sitter spacetime. First, we transform to a different coordinate system

$$r = \sqrt{\frac{3}{\Lambda}} \cosh\left(\sqrt{\frac{\Lambda}{3}} t\right) \sin\chi, \ \tanh\left(\sqrt{\frac{\Lambda}{3}}\tau\right) = \tanh\left(\sqrt{\frac{\Lambda}{3}} t\right) \sec\chi \tag{4.34}$$

where the metric can be written as

$$ds^2 = -\left(1 - \frac{\Lambda r^2}{3}\right) d\tau^2 + \frac{dr^2}{1 - \frac{\Lambda r^2}{3}} + r^2 d\Omega^2 \tag{4.35}$$

where $d\Omega^2$ is now the line element on S^2.

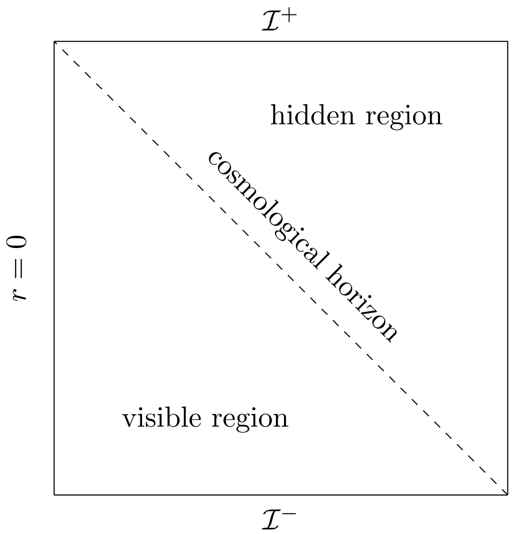

Figure 6: Penrose diagram of 4d De Sitter spacetime.

We can find the Penrose diagram for De Sitter spacetime as shown in fig. 6 where $r = 0$ is the spatial origin of the spacetime. We note that the other timelike boundary is not a real boundary. The spacelike boundaries at past timelike infinity $\mathcal{I}^-$ and future timelike infinity $\mathcal{I}^+$ are real boundaries of the spacetime. This also means that De Sitter spacetime is globally hyperbolic as a constant t slice forms a Cauchy hypersurface of topology S^3. We may now consider an observer stationary at the origin $r = 0$ and find that there is a region of the spacetime hidden to the observer. We also observe the formation of such a cosmological horizon in astronomy when looking to very far away galaxy clusters.
We notice the metric (4.35) has a singularity at $r = \sqrt{3/\Lambda}$. This is a coordinate singularity, similarly to the singularity at $r = 2M$ in Schwarzschild spacetime, and informs us of the existence of an event horizon. The horizon has many similar properties to the event horizon of a black hole. In particular, we may compute its Hawking temperature.

4.2.2 Thermodynamics of De Sitter Spacetime

Since the full derivation of the Hawking temperature involves detailed QFT in curved spacetime (and can be found in Harvey Reall's Black Holes lecture notes), we would like to use a shortcut that was presented as an argument by Hawking himself. Take the definition of the partition function as used in quantum mechanics

$$Z = \mathrm{tr}(e^{-\beta H}) \tag{4.36}$$

where H is the Hamiltonian operator and we are tracing over the Hilbert space of the system. Using thermodynamic language, we also define the Helmholtz free energy F as

$$Z = e^{-\frac{H}{T}} \tag{4.37}$$

where T is the temperature $T = 1/\beta$. Now we also know that in the Schrödinger picture, states evolve as

$$|s(t)\rangle = e^{-iH(t-t_0)} |s(t_0)\rangle \ . \tag{4.38}$$

We notice the partition function looks like the trace over the time evolution operator by $i\beta$. For consistency, we hence require all states in a canonical ensemble with temperature T to be periodic with $i\beta$, i.e. $|s(t)\rangle = |s(t+i\beta)\rangle$. This is known as the KMS condition and we will use it in the opposite direction to argue that the temperature of the spacetime is related to imaginary periodicity.
Consider the horizon of De Sitter spacetime at $r = \sqrt{3/\Lambda}$ with horizon area $A = 12\pi/\Lambda$. The imaginary periodicity motivates us to perform a Wick rotation $\tau \to i\tau_E$ to give the metric

$$ds^2 = \left(1 - \frac{\Lambda r^2}{3}\right) d\tau_E^2 + \frac{dr^2}{1 - \frac{\Lambda r^2}{3}} + r^2 d\Omega^2 \tag{4.39}$$

where we are now looking for a real periodicity in the Euclidean time τ_E. Let's consider the spacetime near the horizon and expand

$$r = \sqrt{\frac{3}{\Lambda}} \left(1 - \frac{1}{2}\lambda\delta^2 + ...\right) \tag{4.40}$$

where λ is a constant and δ is our new coordinate. The metric near the horizon is then

$$ds^2 = \frac{\Lambda}{3}\delta^2 d\tau_E^2 + d\delta^2 + \frac{3}{\Lambda} d\Omega^2 \tag{4.41}$$

where we have chosen $\lambda = \Lambda/3$. We can now see that the first two terms of the metric look like $\mathbb{R}^2$ in polar coordinates where τ_E plays the role of the angular direction. This becomes even more obvious when we define $u = \tau_E\sqrt{3/\Lambda}$ such that this sector of the metric becomes $ds^2 = d\delta^2 + \delta^2 du^2$. To avoid a conical singularity at the origin, we must now identify $u \sim u + 2\pi$ which is the periodicity we have been looking for. The Hawking temperature of De Sitter spacetime is hence

$$T = \frac{1}{2\pi}\sqrt{\frac{\Lambda}{3}} \ . \tag{4.42}$$

Using the value of $\Lambda \approx 1.11 \times 10^{-52}\ m^{-2}$ as determined using data from the Planck satellite, this gives a temperature of $T \approx 2.22 \times 10^{-30}\ K$. Pretty cold.
Now consider the different forms of the partition function

$$Z = e^{-\frac{F}{T}} = \mathrm{tr}(e^{-\beta H}) = \int D[g]\ e^{-\frac{i}{\hbar}S[g]} \tag{4.43}$$

where the Helmholtz free energy is $F = U - TS$. We note that $U = 0$ since there is no matter in the spacetime so $F = -TS$. Now we may compute the action

$$S = \frac{1}{16\pi}\int d^4x\ (R - 2\Lambda) \tag{4.44}$$

since the leading term is classical as $R \neq 2\Lambda$. For this action the classical equation of motion gives $R_{ab} = \Lambda g_{ab}$ and so $R = 4\Lambda$. The action hence evaluates to

$$S = \frac{1}{16\pi}\int d\tau_E r^2 dr \sin\theta d\theta d\phi\ (2i\sqrt{g}\Lambda) = i\frac{3\pi}{\Lambda} \tag{4.45}$$

where we integrate over the compact domain for τ_E and up to the event horizon for r. Picking a suitable normalisation constant for the path integral, we use the saddle point approximation to find

$$Z = e^{\frac{3\pi}{\Lambda\hbar}} = e^{-\frac{F}{T}}\ . \tag{4.46}$$

This allows us to find the Gibbons-Hawking entropy of De Sitter spacetime

$$S = \frac{3\pi}{\Lambda\hbar} = \frac{A}{4\hbar} \tag{4.47}$$

which agrees with Hawking's result for the entropy of any spacetime with horizon area A. Reinstating Newton's constant G, the expression for the entropy is

$$S = \frac{3\pi}{\Lambda G\hbar}\ . \tag{4.48}$$

In the classical and perturbative regime, we take G and $\hbar$ to be small to keep perturbation effects and quantum corrections small. We see that we cannot do this for the entropy. Therefore we have found a truly non-perturbative and intrinsically quantum result. The origin of this entropy is still somewhat of a mystery in general and is thought to stem from quantum gravity. In fact, a full accounting of the entropy of a Schwarzschild black hole from microstates has been achieved in string theory. For De Sitter spacetime, the entropy is believed to reflect the events in the hidden spacetime region. It also gives rise to the problem that the Gibbons-Hawking entropy appears to be independent of the type of matter in the spacetime but the entanglement entropy computed from QFT depends on the number of species. This is known as the species problem. There have been some recent advances in approaching this problem however no full resolution has been found.

4.3 Quantising a Black Hole

We now apply the same method to Schwarzschild spacetime to determine its thermodynamics. The Schwarzschild metric in Schwarzschild coordinates is

$$ds^2 = -\left(1 - \frac{2M}{r}\right) dt^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2 \tag{4.49}$$

where M is a mass parameter (conveniently matching the ADM and Komar mass). We again Wick rotate $t \to i\tau_E$ to obtain the metric

$$ds^2 = \left(1 - \frac{2M}{r}\right) d\tau_E^2 + \frac{dr^2}{1 - \frac{2M}{r}} + r^2 d\Omega^2 \tag{4.50}$$

which has a coordinate singularity at $r = 2M$, corresponding to the event horizon. Expanding $r = 2M + \lambda\delta^2$, the metric near the horizon is

$$ds^2 = \frac{\delta^2}{16M^2} d\tau_E^2 + d\delta^2 + 4M^2 d\Omega^2 \tag{4.51}$$

where we let $\lambda = 1/8M$. To avoid the conical singularity, we identify $\tau_E \sim \tau_E + 8\pi M$ and so

$$T = \frac{1}{8\pi M} \ . \tag{4.52}$$

For a black hole with one solar mass $M \approx 2 \times 10^{30}\ kg$, the temperature is $T \approx 6 \times 10^{-8}\ K$. This is somewhat colder than the surface temperature of the sun so we should not expect to be able to measure it. For a black hole with the mass of an average mountain $M \approx 10^{12}\ K$, the temperature is around $T \approx 10^{12}\ K$. For a black hole of Planck mass $M \approx 10^{-8}\ kg$, we obtain $T \approx 10^{32}\ K$. We see that black holes are very cold for most of their life with their temperature rapidly increasing towards the end of their evaporation. Due to the lower mass bound of the formation of astrophysical black holes, we do not expect any black holes to evaporate anytime soon however there is a question about primordial black holes that formed during first fractions of a second of the universe. These do not necessarily have a lower or upper mass bound since they could have formed through quantum fluctuations. Apart from being candidates for the supermassive black holes at the centre of galaxies, they could evaporate, emitting very high energy particles in the last moments of their life. This could be measured but no such events have been confirmed yet.

To find the entropy of the Euclidean Schwarzschild spacetime (4.50), we have to do a little bit more work. We introduce a radial cut-off at $r = R$ which we later want to push to infinity to obtain the full Schwarzschild solution. This gives rise to a hypersurface of constant r and the action is modified to

$$S = \frac{1}{16\pi} \int d^4x \sqrt{g}\, R + \frac{1}{8\pi} \int d^3x \sqrt{\gamma}\, K \tag{4.53}$$

where γ_{ab} is the induced Riemannian metric on the hypersurface and K_{ab} its associated extrinsic curvature. The unit surface normal is $n^a = (0, \sqrt{1-2M/r}, 0, 0)$. In the asymptotic limit $r \to \infty$, we can then compute the extrinsic curvature scalar

$$K = K^a_{\ a} = \nabla_a n^a = \frac{2}{r} - \frac{2M}{r^2} \tag{4.54}$$

where we can already see the boundary term diverges as we push $R \to \infty$ due to the first term in the extrinsic curvature scalar. Consider flat Euclidean space and repeat the same procedure. We find that the extrinsic curvature scalar only gives the first term $K_0 = 2/r$. We define the action of Minkowski spacetime to give zero and hence subtract off the first term, leaving us with a finite boundary term in the limit $R \to \infty$. We may evaluate the boundary term and substitute into the saddle point approximation of the path integral to obtain $S = 4\pi M^2$ which again matches the horizon area formula.
The Gibbons-Hawking entropy for a black hole together with the Hawking temperature lead to the notion of black hole thermodynamics. For any black hole, we map the temperature to the surface gravity through $T = \kappa/2\pi$ and the entropy to the horizon area through $S = A/4$. This identification allows us to conceptually treat the first and second laws of black hole mechanics (which are theorems) like the first and second laws of thermodynamics. The third law of black hole mechanics, which would map to the third law of thermodynamics, is slightly more controversial and appears to be wrong in the current formulation. Some work remains to be done in black hole thermodynamics.

4.4 Cosmology Rebooted

We have previously looked at classical FLRW spacetime and also quantised De Sitter spacetime. Most cosmological models include matter so we shall do so too. The simplest form of matter is a single massless real scalar field. We again consider the FLRW metric

$$ds^2 = -\alpha^2 dt^2 + a^2(t) d\Omega_3^2 \tag{4.55}$$

on a closed universe with topology $\mathbb{R} \times S^3$. The minisuperspace action now becomes

$$S = \int dt\, \alpha a^3 \left(\frac{6}{a^2} - \frac{6\dot{a}^2}{a^2\alpha^2} + \frac{1}{2\alpha^2}\dot{\phi}^2 \right) \tag{4.56}$$

where we have assumed $\phi = \phi(t)$. The three degrees of freedom then give us three classical equations of motion

$$\alpha:\ 6a + \frac{6a\dot{a}^2}{\alpha^2} - \frac{a^3\dot{\phi}^2}{2\alpha^2} = 0 \tag{4.57}$$

$$\phi:\ a^3\frac{\dot{\phi}}{\alpha} = E = \text{const.} \tag{4.58}$$

$$a:\ \frac{d}{dt}\left(\frac{-12a\dot{a}}{\alpha}\right) - 6\alpha + \frac{6\dot{a}^2}{\alpha} - \frac{3\dot{a}^2\dot{\phi}^2}{2\alpha} = 0 \tag{4.59}$$

where we are now allowed to make a gauge transformation to fix $\alpha = 1$. The total energy density of the scalar field is $E = a^3\dot{\phi}$. The α equation can then be rewritten as

$$\dot{a} = \pm\sqrt{\frac{E^2}{12a^4} - 1} \tag{4.60}$$

where we define $x = \frac{12^{1/4}a}{E^{1/2}}$. The equation of motion is then integrable to give

$$\frac{E}{\sqrt{12}} \int \frac{x^2 dx}{\sqrt{1-x^4}} = \pm(t - t_0) \ . \tag{4.61}$$

We now have a standard integral that can be solved using any good look-up table to give

$$\frac{E}{\sqrt{12}} \frac{1}{2} x^2 \, {}_2F_1\left(\frac{1}{2}, \frac{3}{4}, \frac{7}{4}; x^4\right) = \pm(t - t_0) \tag{4.62}$$

where ${}_2F_1(a, b, c; z)$ is the hypergeometric function. Expanding this in various limits, we find $a \sim t^{1/2}$ when a is small and $\dot{a} = 0$ then $a^4 = E^2/12$. The universe therefore expands rapidly at first, then stops and contracts again. We now use the ϕ equation of motion

$$a^3\dot{\phi} = a^3 \frac{d\phi}{da}\frac{da}{dt} = E \tag{4.63}$$

to find an equation

$$\frac{d\phi}{da} = \pm\frac{E}{a^3}\frac{1}{\sqrt{\frac{E}{12a^4} - 1}} \tag{4.64}$$

for ϕ in terms of a. We again have a standard integral that gives

$$\phi(a) = \pm\frac{1}{\sqrt{3}}\cosh^{-1}\left(\frac{E}{\sqrt{12}a^2}\right) . \tag{4.65}$$

The solutions for $a(t)$ and $\phi(t) = \phi(a(t))$ fully determine the classical evolution of the universe. To quantise this cosmological model, we turn it into a Hamiltonian problem with canonical momenta

$$\pi_\alpha = 0, \ \pi_a = -\frac{12a\dot{a}}{\alpha}, \ \pi_\phi = \frac{a^3\dot{\phi}}{\alpha} \tag{4.66}$$

and Hamiltonian

$$H = \pi_a\dot{a} + \pi_\phi\dot{\phi} + \pi_\alpha\dot{\alpha} - L = -\frac{\pi_a^2}{24a} + \frac{\pi_\phi^2}{2a^3} - 6a \tag{4.67}$$

where the π_a term has the familiar minus sign. We choose the standard representation $\pi_a \to -i\hbar\,\partial/\partial a$ and $\pi_\phi \to -i\hbar\,\partial/\partial\phi$. The wave function ψ now gives the quantum mechanical distribution of both a and ϕ so we require $\psi = \psi(a, \phi)$. The Wheeler-DeWitt equation becomes

$$\left(\frac{1}{24a}\frac{\partial^2}{\partial a^2} - \frac{1}{2a^3}\frac{\partial^2}{\partial\phi^2} - 6a\right)\psi = 0 \ . \tag{4.68}$$

We would like to change our choice of operator ordering to bring the equation into the form

$$\left(\frac{1}{a^2}\frac{\partial}{\partial a}\left(a\frac{\partial}{\partial a}\right) - \frac{12}{a^3}\frac{\partial^2}{\partial\phi^2} - 144a\right)\psi = 0 \tag{4.69}$$

which has the benefit that now we can perform a change of coordinate $a = e^y$ to put the singularity at $y \to -\infty$. The equation turns into the more convenient form

$$\left(\frac{\partial^2}{\partial y^2} - 12\frac{\partial^2}{\partial \phi^2} - 144e^y\right)\psi = 0 \tag{4.70}$$

which can be solved using the ansatz $\psi_k(y, \phi) = f(y)e^{ik\phi}$ where $f(y) = K_{\sqrt{3}ik}(\sqrt{48}e^{2y})$ where K is a modified Bessel function. An arbitrary state is a superposition of these modes

$$\psi = \int dk\ \tilde{f}(k)\psi_k(y, \phi) \tag{4.71}$$

where we can choose $\tilde{f}(k)$ to specify a particular state. Choosing

$$\tilde{f}(k) = e^{-\frac{(k-\tilde{k})^2}{2b^2}} \tag{4.72}$$

with the scalar field peaked at $\tilde{k}$ corresponds to a particular choice of E. This returns a wave function

$$\psi \sim e^{-\frac{1}{2}b^2\left(\phi \pm \frac{1}{\sqrt{3}}\cosh^{-1}\left(\frac{E}{\sqrt{12}a^2}\right)\right)^2} \times \text{phase} \tag{4.73}$$

peaked at the classical solution.

4.5 Behaviour of Singularities

In the previous chapter, we considered the Kasner universe that is believed to describe spacetime near a singularity. We would like to follow the reasoning of Belinski, Khalatnikov and Lifshitz to understand singularities in more detail.

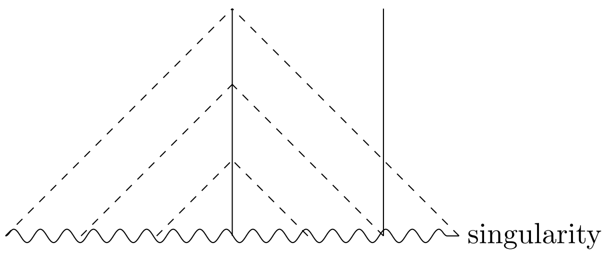

Figure 7: Past lights cones near the singularity.

The first interesting effect is that near a singularity, points become independent from each other. This is shown in fig. 7 where we consider two trajectories and see that as we get closer and closer to the singularity, the past light cones do not overlap. Very close to the singularity we can treat each point in space by itself. Setting $\Lambda = 0$, the Hamiltonian takes the form

$$H = (\text{kinetic term}) + (\text{potential term}) \tag{4.74}$$

where the potential term includes terms like $(\gamma^{-1}\partial\gamma)^2$ and $(\gamma^{-1}\partial^2\gamma)$ that become irrelevant when the points in space are independent from each other. At each point in spacetime we now consider the Kasner metric

$$ds^2 = -dt^2 + t^{2p_1}dx^2 + t^{2p_2}dy^2 + t^{2p_3}dz^2 \tag{4.75}$$

satisfying the vacuum Einstein equation $p_1 + p_2 + p_3 = 1$ and $p_1^2 + p_2^2 + p_3^2 = 1$. The two conditions can be solved to describe the exponents with a single variable u

$$p_1 = \frac{-u}{1+u+u^2},\ p_2 = \frac{1+u}{1+u+u^2},\ p_1 = \frac{u(u+1)}{1+u+u^2} \tag{4.76}$$

where $0 \leq u \leq 1$. We have made the choice that $p_1 < 0$ and $0 < p_2 < p_3$ here without loss of generality. As a similar choice to cosmology, we change our coordinates to

$$t^{2p_i} = e^{-2\beta_i} \tag{4.77}$$

where $\beta_i = v_i\tau + \beta_i^{(0)}$. The coordinate τ is the new time coordinate and v_i can be interpreted as the velocity of expansion. We can find these in terms of the old variables as

$$p_i = \frac{v_i}{v_1 + v_2 + v_3},\ \tau = \frac{-\ln t}{v_1 + v_2 + v_3} + \text{const.} \tag{4.78}$$

putting the $t = 0$ singularity at $\tau \to \infty$. The benefit of this change of variables is that the Hamiltonian constraint can be written as

$$\dot{\beta}_1\dot{\beta}_2 + \dot{\beta}_2\dot{\beta}_3 + \dot{\beta}_3\dot{\beta}_1 = 0 \tag{4.79}$$

which is equivalent to a null trajectory $g_{ij}\dot{\beta}_i\dot{\beta}_j = 0$ through a spacetime with metric

$$g_{ij} = \begin{pmatrix} 0 & -1 & -1 \\ -1 & 0 & -1 \\ -1 & -1 & 0 \end{pmatrix}. \tag{4.80}$$

By computing Hamilton's equation

$$\frac{d\gamma_{ij}}{dt} = \{\gamma_{ij}, H\}, \tag{4.81}$$

it is possible to show that this trajectory is a geodesic. We have hence shown that the system traced out a null geodesic through minisuperspace with a flat metric of signature $(-,+,+)$. To see this more clearly, we may change coordinates again to

$$\beta^1 = \frac{1}{\sqrt{6}}(T - X - \sqrt{3}Y),\ \beta^2 = \frac{1}{\sqrt{6}}(T - X + \sqrt{3}Y),\ \beta^3 = \frac{1}{\sqrt{6}}(T + 2X) \tag{4.82}$$

where the metric is now exactly 3d Minkowski with line element $ds^2 = -dT^2 + dX^2 + dY^2$. Hence the BKL idealisation makes superspace a copy of 3d Minkowski spacetime at each point in space.

Now let's get back to the Kasner metric. We reassign

$$t^{p_1} = e^{\alpha},\ t^{p_2} = e^{\beta},\ t^{p_3} = e^{\gamma} \tag{4.83}$$

and consider the Hamiltonian $H = T + V$. BKL argue that the potential term V introduces rapid changes in the u from time to time. We now also allow u to leave the range between 0 and 1 such that p_1, p_2 and p_3 can have an arbitrary order. In a type I change, $u \to -u-1$ such that p_1 and p_2 swap. In a type II change, $u \to 1/u$ such that p_2 and p_3 swap.

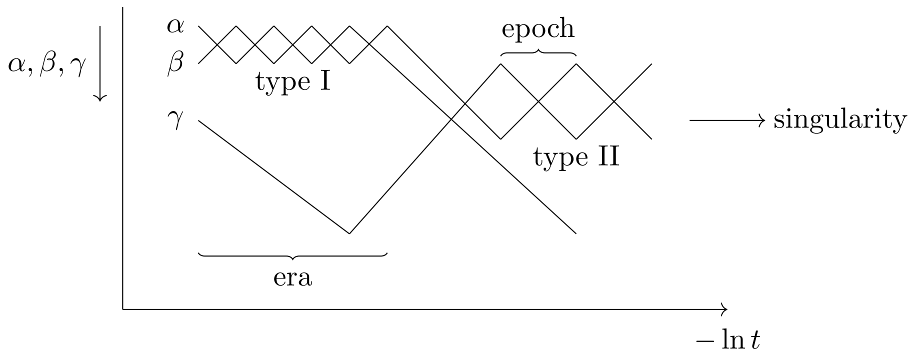

Figure 8: Behaviour of metric coefficients close to the singularity.

This behaviour is displayed in fig. 8 where we meet chaotic oscillations as we get close to the singularity on the right. The period between each swap is called an epoch and the period where only one type of swap is active is called an era. The evolution of α, β and γ reminds of null geodesics bouncing off walls.
To analyse this behaviour more, we switch to polar coordinates in superspace

$$-dT^2 + dR^2 + R^2 d\phi^2 \tag{4.84}$$

and define a new set of coordinates $R = w \sinh \rho$ and $T = w \cosh \rho$. The metric is now

$$-dw^2 + w^2(d\rho^2 + \sinh^2 \rho d\phi^2) \tag{4.85}$$

where we identify the term in brackets $d\rho^2 + \sinh^2 \rho d\phi^2$ with Lobachevsky space. We hence change our coordinates on this subspace yet again

$$u = \frac{\sinh \rho \cos \phi}{\cosh \rho - \sinh \rho \sin \phi}, \; v = \frac{1}{\cosh \rho - \sinh \rho \sin \phi} \tag{4.86}$$

such that the metric simplies to

$$\frac{du^2 + dv^2}{v^2}, \; v \geq 0 \; . \tag{4.87}$$

In $u - v$ space, we can now easily display the evolution of the metric as bouncing off walls.

Fig. 9 describes the $u - v$ plane of Lobachevsky space. We can describe our system as a particle confined to the region bounded by the walls A, B and C. Introducing a complexified coordinate $z = u + iv$, geodesics in this space are described by circles $|z - u_0|^2 = c^2$ where u_0 is the origin of the circle and c the radius. Bouncing off the walls now generated maps

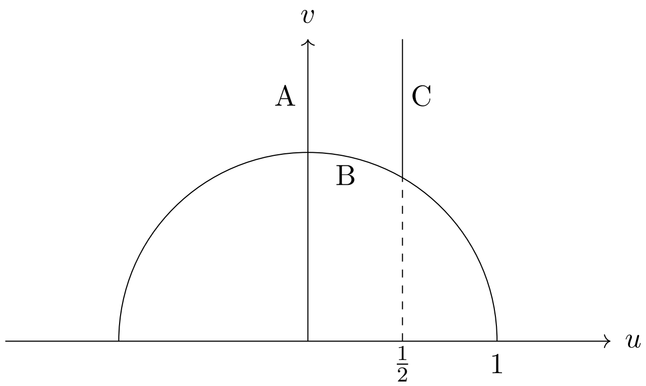

Figure 9: Relevant region of Lobachevsky space.

$A : z \to -\bar{z}$, $B : z \to 1 - \bar{z}$ and $C : z \to 1/\bar{z}$ on the coordinate z where $\bar{z}$ denotes the complex conjugate. We recognise the region bounded by A, B and C to be the fundamental region of the integer matrix group $GL(2, \mathbb{Z})$ and the transformations corresponding to A, B and C to be generators in $GL(2, \mathbb{Z})$. Therefore we may consider the trajectory of the spacetime as a word in $GL(2, \mathbb{Z})$.
Quantising the spacetime gives the Wheeler-DeWitt equation

$$\left(-\frac{\partial^2}{\partial w^2} - \frac{2}{w}\frac{\partial}{\partial w} + \frac{1}{w^2}\Delta_L\right)\psi = 0 \tag{4.88}$$

where Δ_L is the Laplacian on Lobachevsky space. We have to also implement the boundary conditions $\psi = 0$ on the walls which leads to the usual effects of containing a wave function. Solving this equation can be done by seeking separable solutions of the form

$$\psi = g(w)f(u, v) \tag{4.89}$$

and finding the eigenfunctions f of the Lobachevsky Laplacian such that $\Delta_L f = -E_n f$. We then find solutions for g of the form

$$g(w) = w^p,\ p = \frac{-1 \pm \sqrt{1 - 4E_n}}{2} = -\frac{1}{2} + it_n\ . \tag{4.90}$$